

TENSORS AND DIFFERENTIAL FORMS

4.1 TENSOR ANALYSIS

Introduction, Properties

Tensors are important in many areas of physics, ranging from topics such as general relativity and electrodynamics to descriptions of the properties of bulk matter such as stress (the pattern of force applied to a sample) and strain (its response to the force), or the moment of inertia (the relation between a torsional force applied to an object and its resultant angular acceleration). Tensors constitute a generalization of quantities previously introduced: scalars and vectors. We identified a **scalar** as an quantity that remained invariant under rotations of the coordinate system and which could be specified by the value of a single real number. **Vectors** were identified as quantities that had a number of real components equal to the dimension of the coordinate system, with the components transforming like the coordinates of a fixed point when a coordinate system is rotated. Calling scalars **tensors of rank 0** and vectors **tensors of rank 1**, we identify a tensor of rank n in a d -dimensional space as an object with the following properties:

- It has components labeled by n indices, with each index assigned values from 1 through d , and therefore having a total of d^n components;
- The components transform in a specified manner under coordinate transformations.

The behavior under coordinate transformation is of central importance for tensor analysis and conforms both with the way in which mathematicians define linear spaces and

with the physicist's notion that physical observables must not depend on the choice of coordinate frames.

Covariant and Contravariant Tensors

In Chapter 3, we considered the rotational transformation of a vector $\mathbf{A} = A_1 \hat{\mathbf{e}}_1 + A_2 \hat{\mathbf{e}}_2 + A_3 \hat{\mathbf{e}}_3$ from the Cartesian system defined by $\hat{\mathbf{e}}_i$ ($i = 1, 2, 3$) into a rotated coordinate system defined by $\hat{\mathbf{e}}'_i$, with the same vector $\mathbf{A}$ then represented as $\mathbf{A}' = A'_1 \hat{\mathbf{e}}'_1 + A'_2 \hat{\mathbf{e}}'_2 + A'_3 \hat{\mathbf{e}}'_3$. The components of $\mathbf{A}$ and $\mathbf{A}'$ are related by

$$A'_i = \sum_j (\hat{\mathbf{e}}'_i \cdot \hat{\mathbf{e}}_j) A_j, \quad (4.1)$$

where the coefficients $(\hat{\mathbf{e}}'_i \cdot \hat{\mathbf{e}}_j)$ are the projections of $\hat{\mathbf{e}}'_i$ in the $\hat{\mathbf{e}}_j$ directions. Because the $\hat{\mathbf{e}}_i$ and the $\hat{\mathbf{e}}_j$ are linearly related, we can also write

$$A'_i = \sum_j \frac{\partial x'_i}{\partial x_j} A_j. \quad (4.2)$$

The formula of Eq. (4.2) corresponds to the application of the chain rule to convert the set A_j into the set A'_i , and is valid for A_j and A'_i of arbitrary magnitude because both vectors depend linearly on their components.

We have also previously noted that the gradient of a scalar φ has in the unrotated Cartesian coordinates the components $(\nabla\varphi)_j = (\partial\varphi/\partial x_j)\hat{\mathbf{e}}_j$, meaning that in a rotated system we would have

$$(\nabla\varphi)'_i \equiv \frac{\partial\varphi}{\partial x'_i} = \sum_j \frac{\partial x_j}{\partial x'_i} \frac{\partial\varphi}{\partial x_j}, \quad (4.3)$$

showing that the gradient has a transformation law that differs from that of Eq. (4.2) in that $\partial x'_i/\partial x_j$ has been replaced by $\partial x_j/\partial x'_i$. Remembering that these two expressions, if written in detail, correspond, respectively, to $(\partial x'_i/\partial x_j)_{x_k}$ and $(\partial x_j/\partial x'_i)_{x'_k}$, where k runs over the index values other than that already in the denominator, and also noting that (in Cartesian coordinates) they are two different ways of computing the same quantity (the magnitude and sign of the projection of one of these unit vectors upon the other), we see that it was legitimate to identify both $\mathbf{A}$ and $\nabla\varphi$ as *vectors*, as we did in Chapter 3.

However, as the alert reader may note from the repeated insertion of the word “Cartesian,” the partial derivatives in Eqs. (4.2) and (4.3) are only guaranteed to be equal in Cartesian coordinate systems, and since there is sometimes a need to use non-Cartesian systems it becomes necessary to distinguish these two different transformation rules. Quantities transforming according to Eq. (4.2) are called **contravariant** vectors, while those transforming according to Eq. (4.3) are termed **covariant**. When non-Cartesian systems may be in play, it is therefore customary to distinguish these transformation properties by writing the index of a contravariant vector as a superscript and that of a covariant vector as a subscript. This means, among other things, that the components of the position vector $\mathbf{r}$,

which is contravariant, must now be written (x^1, x^2, x^3) . Thus, summarizing,

$$(A')^i = \sum_j \frac{\partial(x')^i}{\partial x^j} A^j \quad \mathbf{A}, \text{ a contravariant vector}, \quad (4.4)$$

$$A'_i = \sum_j \frac{\partial x^j}{\partial(x')^i} A_j \quad \mathbf{A}, \text{ a covariant vector}. \quad (4.5)$$

It is useful to note that the occurrence of subscripts and superscripts is systematic; the **free** (i.e., unsummed) index i occurs as a superscript on both sides of Eq. (4.4), while it appears as a subscript on both sides of Eq. (4.5), if we interpret an upper index in the denominator as equivalent to a lower index. The summed index occurs once as upper and once as lower (again treating an upper index in the denominator as a lower index). A frequently used shorthand (the **Einstein convention**) is to omit the summation sign in formulas like Eqs. (4.4) and (4.5) and to understand that when the same symbol occurs both as an upper and a lower index in the same expression, it is to be summed. We will gradually back into the use of the Einstein convention, giving the reader warnings as we start to do so.

Tensors of Rank 2

Now we proceed to define **contravariant, mixed, and covariant tensors of rank 2** by the following equations for their components under coordinate transformations:

$$\begin{aligned} (A')^{ij} &= \sum_{kl} \frac{\partial(x')^i}{\partial x^k} \frac{\partial(x')^j}{\partial x^l} A^{kl}, \\ (B')^i_j &= \sum_{kl} \frac{\partial(x')^i}{\partial x^k} \frac{\partial x^l}{\partial(x')^j} B_l^k, \\ (C')_{ij} &= \sum_{kl} \frac{\partial x^k}{\partial(x')^i} \frac{\partial x^l}{\partial(x')^j} C_{kl}. \end{aligned} \quad (4.6)$$

Clearly, the rank goes as the number of partial derivatives (or direction cosines) in the definition: 0 for a scalar, 1 for a vector, 2 for a second-rank tensor, and so on. Each index (subscript or superscript) ranges over the number of dimensions of the space. The number of indices (equal to the rank of tensor) is not limited by the dimensionality of the space. We see that A^{kl} is contravariant with respect to both indices, C_{kl} is covariant with respect to both indices, and B_l^k transforms contravariantly with respect to the index k but covariantly with respect to the index l . Once again, if we are using Cartesian coordinates, all three forms of the tensors of second rank, contravariant, mixed, and covariant are the same.

As with the components of a vector, the transformation laws for the components of a tensor, Eq. (4.6), cause its physically relevant properties to be independent of the choice of reference frame. This is what makes tensor analysis important in physics. The independence relative to reference frame (invariance) is ideal for expressing and investigating universal physical laws.

The second-rank tensor A (with components A^{kl}) may be conveniently represented by writing out its components in a square array (3×3 if we are in three-dimensional (3-D) space):

$$A = \begin{pmatrix} A^{11} & A^{12} & A^{13} \\ A^{21} & A^{22} & A^{23} \\ A^{31} & A^{32} & A^{33} \end{pmatrix}. \quad (4.7)$$

This does not mean that any square array of numbers or functions forms a tensor. The essential condition is that the components transform according to Eq. (4.6).

We can view each of Eq. (4.6) as a matrix equation. For A , it takes the form

$$(A')^{ij} = \sum_{kl} S_{ik} A^{kl} (S^T)_{lj}, \quad \text{or} \quad A' = SAS^T, \quad (4.8)$$

a construction that is known as a **similarity transformation** and is discussed in Section 5.6.

In summary, tensors are systems of components organized by one or more indices that transform according to specific rules under a set of transformations. The number of indices is called the rank of the tensor.

Addition and Subtraction of Tensors

The addition and subtraction of tensors is defined in terms of the individual elements, just as for vectors. If

$$A + B = C, \quad (4.9)$$

then, taking as an example A , B , and C to be contravariant tensors of rank 2,

$$A^{ij} + B^{ij} = C^{ij}. \quad (4.10)$$

In general, of course, A and B must be tensors of the same rank (of both contra- and co-variance) and in the same space.

Symmetry

The order in which the indices appear in our description of a tensor is important. In general, A^{mn} is independent of A^{nm} , but there are some cases of special interest. If, for all m and n ,

$$A^{mn} = A^{nm}, \quad A \text{ is symmetric.} \quad (4.11)$$

If, on the other hand,

$$A^{mn} = -A^{nm}, \quad A \text{ is antisymmetric.} \quad (4.12)$$

Clearly, every (second-rank) tensor can be resolved into symmetric and antisymmetric parts by the identity

$$A^{mn} = \frac{1}{2}(A^{mn} + A^{nm}) + \frac{1}{2}(A^{mn} - A^{nm}), \quad (4.13)$$

the first term on the right being a symmetric tensor, the second, an antisymmetric tensor.

Isotropic Tensors

To illustrate some of the techniques of tensor analysis, let us show that the now-familiar Kronecker delta, δ_{kl} , is really a mixed tensor of rank 2, δ_l^k .¹ The question is: Does δ_l^k transform according to Eq. (4.6)? This is our criterion for calling it a tensor. If δ_l^k is the mixed tensor corresponding to this notation, it must satisfy (using the summation convention, meaning that the indices k and l are to be summed)

$$(\delta')_j^i = \frac{\partial(x')^i}{\partial x^k} \frac{\partial x^l}{\partial(x')^j} \delta_l^k = \frac{\partial(x')^i}{\partial x^k} \frac{\partial x^k}{\partial(x')^j},$$

where we have performed the l sum and used the definition of the Kronecker delta. Next,

$$\frac{\partial(x')^i}{\partial x^k} \frac{\partial x^k}{\partial(x')^j} = \frac{\partial(x')^i}{\partial(x')^j},$$

where we have identified the k summation on the left-hand side as an instance of the chain rule for differentiation. However, $(x')^i$ and $(x')^j$ are independent coordinates, and therefore the variation of one with respect to the other must be zero if they are different, unity if they coincide; that is,

$$\frac{\partial(x')^i}{\partial(x')^j} = (\delta')_j^i. \quad (4.14)$$

Hence

$$(\delta')_j^i = \frac{\partial(x')^i}{\partial x^k} \frac{\partial x^l}{\partial(x')^j} \delta_l^k, \quad (4.15)$$

showing that the δ_l^k are indeed the components of a mixed second-rank tensor. Note that this result is independent of the number of dimensions of our space.

The Kronecker delta has one further interesting property. It has the same components in all of our rotated coordinate systems and is therefore called **isotropic**. In Section 4.2 and Exercise 4.2.4 we shall meet a third-rank isotropic tensor and three fourth-rank isotropic tensors. No isotropic first-rank tensor (vector) exists.

Contraction

When dealing with vectors, we formed a scalar product by summing products of corresponding components:

$$\mathbf{A} \cdot \mathbf{B} = \sum_i A_i B_i.$$

The generalization of this expression in tensor analysis is a process known as contraction. Two indices, one covariant and the other contravariant, are set equal to each other, and then (as implied by the summation convention) we sum over this repeated index. For example,

¹It is common practice to refer to a tensor $\mathbf{A}$ by specifying a typical component, such as A_{ij} , thereby also conveying information as to its covariant vs. contravariant nature. As long as you refrain from writing nonsense such as $\mathbf{A} = A_{ij}$, no harm is done.

let us contract the second-rank mixed tensor B_j^i by setting j to i , then summing over i . To see what happens, let's look at the transformation formula that converts B into B' . Using the summation convention,

$$(B')_i^i = \frac{\partial(x')^i}{\partial x^k} \frac{\partial x^l}{\partial(x')^i} B_l^k = \frac{\partial x^l}{\partial x^k} B_l^k,$$

where we recognized the i summation as an instance of the chain rule for differentiation. Then, because the x^i are independent, we may use Eq. (4.14) to reach

$$(B')_i^i = \delta_k^l B_l^k = B_k^k. \quad (4.16)$$

Remembering that the repeated index (i or k) is summed, we see that the contracted B is invariant under transformation and is therefore a scalar.² In general, the operation of contraction reduces the rank of a tensor by 2.

Direct Product

The components of two tensors (of any ranks and covariant/contravariant characters) can be multiplied, component by component, to make an object with all the indices of both factors. The new quantity, termed the **direct product** of the two tensors, can be shown to be a tensor whose rank is the sum of the ranks of the factors, and with covariant/contravariant character that is the sum of those of the factors. We illustrate:

$$C_{klm}^{ij} = A_k^i B_{lm}^j, \quad F_{kl}^{ij} = A^j B_{lk}^i.$$

Note that the index order in the direct product can be defined as desired, but the covariance/contravariance of the factors must be maintained in the direct product.

Example 4.1.1 DIRECT PRODUCT OF TWO VECTORS

Let's form the direct product of a covariant vector a_i (rank-1 tensor) and a contravariant vector b^j (also a rank-1 tensor) to form a mixed tensor of rank 2, with components $C_i^j = a_i b^j$. To verify that C_i^j is a tensor, we consider what happens to it under transformation:

$$(C')_i^j = (a')_i (b')^j = \frac{\partial x^k}{\partial(x')^i} a_k \frac{\partial(x')^j}{\partial x^l} b_l = \frac{\partial x^k}{\partial(x')^i} \frac{\partial(x')^j}{\partial x^l} C_k^l, \quad (4.17)$$

confirming that C_i^j is the mixed tensor indicated by its notation.

If we now form the contraction C_i^i (remember that i is summed), we obtain the scalar product $a_i b^i$. From Eq. (4.17) it is easy to see that $a_i b^i = (a')_i (b')^i$, indicating the invariance required of a scalar product. ■

Note that the direct product concept gives a meaning to quantities such as $\nabla \mathbf{E}$, which was not defined within the framework of vector analysis. However, this and other tensor-like quantities involving differential operators must be used with caution, because their

²In matrix analysis this scalar is the **trace** of the matrix whose elements are the B_j^i .

transformation rules are simple only in Cartesian coordinate systems. In non-Cartesian systems, operators $\partial/\partial x^i$ act also on the partial derivatives in the transformation expressions and alter the tensor transformation rules.

We summarize the key idea of this subsection:

The direct product is a technique for creating new, higher-rank tensors.

Inverse Transformation

If we have a contravariant vector A^i , which must have the transformation rule (using summation convention)

$$(A')^j = \frac{\partial(x')^j}{\partial x^i} A^i,$$

the inverse transformation (which can be obtained simply by interchanging the roles of the primed and unprimed quantities) is

$$A^i = \frac{\partial x^i}{\partial(x')^j} (A')^j, \quad (4.18)$$

as may also be verified by applying $\partial(x')^k/\partial x^i$ (and summing i) to A^i as given by Eq. (4.18):

$$\frac{\partial(x')^k}{\partial x^i} A^i = \frac{\partial(x')^k}{\partial x^i} \frac{\partial x^i}{\partial(x')^j} (A')^j = \delta_j^k (A')^j = (A')^k. \quad (4.19)$$

We see that $(A')^k$ is recovered. Incidentally, note that

$$\frac{\partial x^i}{\partial(x')^j} \neq \left[\frac{\partial(x')^j}{\partial x^i} \right]^{-1};$$

as we have previously pointed out, these derivatives have different other variables held fixed. The cancellation in Eq. (4.19) only occurs because the product of derivatives is summed. In Cartesian systems, we do have

$$\frac{\partial x^i}{\partial(x')^j} = \frac{\partial(x')^j}{\partial x^i},$$

both equal to the direction cosine connecting the x^i and $(x')^j$ axes, but this equality does not extend to non-Cartesian systems.

Quotient Rule

If, for example, A_{ij} and B_{kl} are tensors, we have already observed that their direct product, $A_{ij}B_{kl}$, is also a tensor. Here we are concerned with the inverse problem, illustrated by

equations such as

$$\begin{aligned}
 K_i A^i &= B, \\
 K_i^j A_j &= B_i, \\
 K_i^j A_{jk} &= B_{ik}, \\
 K_{ijkl} A^{ij} &= B_{kl}, \\
 K^{ij} A^k &= B^{ijk}.
 \end{aligned} \tag{4.20}$$

In each of these expressions A and B are known tensors of ranks indicated by the number of indices, A is arbitrary, and the summation convention is in use. In each case K is an unknown quantity. We wish to establish the transformation properties of K. The quotient rule asserts:

If the equation of interest holds in all transformed coordinate systems, then K is a tensor of the indicated rank and covariant/contravariant character.

Part of the importance of this rule in physical theory is that it can establish the tensor nature of quantities. For example, the equation giving the dipole moment **m** induced in an anisotropic medium by an electric field **E** is

$$m_i = P_{ij} E^j.$$

Since presumably we know that **m** and **E** are vectors, the general validity of this equation tells us that the **polarization matrix** P is a tensor of rank 2.

Let's prove the quotient rule for a typical case, which we choose to be the second of Eqs. (4.20). If we apply a transformation to that equation, we have

$$K_i^j A_j = B_i \quad \longrightarrow \quad (K')_i^j A'_j = B'_i. \tag{4.21}$$

We now evaluate B'_i , reaching the last member of the equation below by using Eq. (4.18) to convert A_j into components of A' (note that this is the **inverse** of the transformation to the primed quantities):

$$B'_i = \frac{\partial x^m}{\partial (x')^i} B_m = \frac{\partial x^m}{\partial (x')^i} K_m^j A_j = \frac{\partial x^m}{\partial (x')^i} K_m^j \frac{\partial (x')^n}{\partial x^j} A'_n. \tag{4.22}$$

It may lessen possible confusion if we rename the dummy indices in Eq. (4.22), so we interchange n and j , causing that equation to then read

$$B'_i = \frac{\partial x^m}{\partial (x')^i} \frac{\partial (x')^j}{\partial x^n} K_m^n A'_j. \tag{4.23}$$

It has now become clear that if we subtract the expression for B'_i in Eq. (4.23) from that in Eq. (4.21) we will get

$$\left[(K')_i^j - \frac{\partial x^m}{\partial (x')^i} \frac{\partial (x')^j}{\partial x^n} K_m^n \right] A'_j = 0. \tag{4.24}$$

Since A' is arbitrary, the coefficient of A'_j in Eq. (4.24) must vanish, showing that K has the transformation properties of the tensor corresponding to its index configuration.

Other cases may be treated similarly. One minor pitfall should be noted: The quotient rule does not necessarily apply if $\mathbf{B}$ is zero. The transformation properties of zero are indeterminate.

Example 4.1.2 EQUATIONS OF MOTION AND FIELD EQUATIONS

In classical mechanics, Newton's equations of motion $m\dot{\mathbf{v}} = \mathbf{F}$ tell us on the basis of the quotient rule that, if the mass is a scalar and the force a vector, then the acceleration $\mathbf{a} \equiv \dot{\mathbf{v}}$ is a vector. In other words, the vector character of the force as the driving term imposes its vector character on the acceleration, provided the scale factor m is scalar.

The wave equation of electrodynamics can be written in relativistic four-vector form as

$$\left[\frac{1}{c^2} \frac{\partial^2}{\partial t^2} - \nabla^2 \right] A^\mu = J^\mu,$$

where J^μ is the external charge/current density (a four-vector) and A^μ is the four-component vector potential. The second-derivative expression in square brackets can be shown to be a scalar. From the quotient rule, we may then infer that A^μ must be a tensor of rank 1, i.e., also a four-vector. ■

The quotient rule is a substitute for the illegal division of tensors.

Spinors

It was once thought that the system of scalars, vectors, tensors (second-rank), and so on formed a complete mathematical system, one that is adequate for describing a physics independent of the choice of reference frame. But the universe and mathematical physics are not that simple. In the realm of elementary particles, for example, spin-zero particles³ (π mesons, α particles) may be described with scalars, spin 1 particles (deuterons) by vectors, and spin 2 particles (gravitons) by tensors. This listing omits the most common particles: electrons, protons, and neutrons, all with spin $\frac{1}{2}$. These particles are properly described by **spinors**. A spinor does not have the properties under rotation consistent with being a scalar, vector, or tensor of any rank. A brief introduction to spinors in the context of group theory appears in Chapter 17.

Exercises

- 4.1.1** Show that if all the components of any tensor of any rank vanish in one particular coordinate system, they vanish in all coordinate systems.
Note. This point takes on special importance in the four-dimensional (4-D) curved space of general relativity. If a quantity, expressed as a tensor, exists in one coordinate system, it exists in all coordinate systems and is not just a consequence of a **choice** of a coordinate system (as are centrifugal and Coriolis forces in Newtonian mechanics).

³The particle spin is intrinsic angular momentum (in units of $\hbar$). It is distinct from classical (often called **orbital**) angular momentum that arises from the motion of the particle.

- 4.1.2** The components of tensor A are equal to the corresponding components of tensor B in one particular coordinate system denoted, by the superscript 0; that is,

$$A_{ij}^0 = B_{ij}^0.$$

Show that tensor A is equal to tensor B , $A_{ij} = B_{ij}$, in all coordinate systems.

- 4.1.3** The last three components of a 4-D vector vanish in each of two reference frames. If the second reference frame is not merely a rotation of the first about the x_0 axis, meaning that at least one of the coefficients $\partial(x')^i/\partial x^0$ ($i = 1, 2, 3$) is nonzero, show that the zeroth component vanishes in all reference frames. Translated into relativistic mechanics, this means that if momentum is conserved in two Lorentz frames, then energy is conserved in all Lorentz frames.
- 4.1.4** From an analysis of the behavior of a general second-rank tensor under 90° and 180° rotations about the coordinate axes, show that an isotropic second-rank tensor in 3-D space must be a multiple of δ_j^i .

- 4.1.5** The 4-D fourth-rank Riemann-Christoffel curvature tensor of general relativity, R_{iklm} , satisfies the symmetry relations

$$R_{iklm} = -R_{ikml} = -R_{kilm}.$$

With the indices running from 0 to 3, show that the number of independent components is reduced from 256 to 36 and that the condition

$$R_{iklm} = R_{lmik}$$

further reduces the number of independent components to 21. Finally, if the components satisfy an identity $R_{iklm} + R_{ilmk} + R_{imkl} = 0$, show that the number of independent components is reduced to 20.

Note. The final three-term identity furnishes new information only if all four indices are different.

- 4.1.6** T_{iklm} is antisymmetric with respect to all pairs of indices. How many independent components has it (in 3-D space)?
- 4.1.7** If $T_{...i}$ is a tensor of rank n , show that $\partial T_{...i}/\partial x^j$ is a tensor of rank $n + 1$ (Cartesian coordinates).
Note. In non-Cartesian coordinate systems the coefficients a_{ij} are, in general, functions of the coordinates, and the derivatives the components of a tensor of rank n do not form a tensor except in the special case $n = 0$. In this case the derivative does yield a covariant vector (tensor of rank 1).
- 4.1.8** If $T_{ijk...}$ is a tensor of rank n , show that $\sum_j \partial T_{ijk...}/\partial x^j$ is a tensor of rank $n - 1$ (Cartesian coordinates).
- 4.1.9** The operator

$$\nabla^2 - \frac{1}{c^2} \frac{\partial^2}{\partial t^2}$$

may be written as

$$\sum_{i=1}^4 \frac{\partial^2}{\partial x_i^2},$$

using $x_4 = ict$. This is the 4-D Laplacian, sometimes called the d'Alembertian and denoted by $\square^2$. Show that it is a **scalar** operator, that is, invariant under Lorentz transformations, i.e., under rotations in the space of vectors (x^1, x^2, x^3, x^4) .

- 4.1.10** The double summation $K_{ij}A^iB^j$ is invariant for any two vectors A^i and B^j . Prove that K_{ij} is a second-rank tensor.

Note. In the form ds^2 (invariant) $= g_{ij}dx^i dx^j$, this result shows that the matrix g_{ij} is a tensor.

- 4.1.11** The equation $K_{ij}A^{jk} = B_i^k$ holds for all orientations of the coordinate system. If A and B are arbitrary second-rank tensors, show that K is a second-rank tensor also.

4.2 PSEUDOTENSORS, DUAL TENSORS

The topics of this section will be treated for tensors restricted for practical reasons to Cartesian coordinate systems. This restriction is not conceptually necessary but simplifies the discussion and makes the essential points easy to identify.

Pseudotensors

So far the coordinate transformations in this chapter have been restricted to **passive rotations**, by which we mean rotation of the coordinate system, keeping vectors and tensors at fixed orientations. We now consider the effect of reflections or inversions of the coordinate system (sometimes also called **improper rotations**).

In Section 3.3, where attention was restricted to orthogonal systems of Cartesian coordinates, we saw that the effect of a coordinate rotation on a fixed vector could be described by a transformation of its components according to the formula

$$A' = SA, \quad (4.25)$$

where S was an orthogonal matrix with determinant +1. If the coordinate transformation included a reflection (or inversion), the transformation matrix was still orthogonal, but had determinant -1 . While the transformation rule of Eq. (4.25) was obeyed by vectors describing quantities such as position in space or velocity, it produced the wrong sign when vectors describing angular velocity, torque, and angular momentum were subject to improper rotations. These quantities, called **axial vectors**, or nowadays **pseudovectors**, obeyed the transformation rule

$$A' = \det(S)SA \quad (\text{pseudovector}). \quad (4.26)$$

The extension of this concept to tensors is straightforward. We insist that the designation *tensor* refer to objects that transform as in Eq. (4.6) and its generalization to arbitrary

rank, but we also accommodate the possibility of having, at arbitrary rank, objects whose transformation requires an additional sign factor to adjust for the effect associated with improper rotations. These objects are called **pseudotensors**, and constitute a generalization of the objects already identified as pseudoscalars and pseudovectors.

If we form a tensor or pseudotensor as a direct product or identify one via the quotient rule, we can determine its pseudo status by what amounts to a sign rule. Letting T be a tensor and P a pseudotensor, then, symbolically,

$$T \otimes T = P \otimes P = T, \quad T \otimes P = P \otimes T = P. \quad (4.27)$$

Example 4.2.1 LEVI-CIVITA SYMBOL

The three-index version of the Levi-Civita symbol, introduced in Eq. (2.8), has the values

$$\begin{aligned} \varepsilon_{123} &= \varepsilon_{231} = \varepsilon_{312} = +1, \\ \varepsilon_{132} &= \varepsilon_{213} = \varepsilon_{321} = -1, \\ \text{all other } \varepsilon_{ijk} &= 0. \end{aligned} \quad (4.28)$$

Suppose now that we have a rank-3 pseudotensor η_{ijk} , which in one particular Cartesian coordinate system is equal to ε_{ijk} . Then, letting A stand for the matrix of coefficients in an orthogonal transformation of $\mathbb{R}^3$, we have in the transformed coordinate system

$$\eta'_{ijk} = \det(A) \sum_{pqr} a_{ip} a_{jq} a_{kr} \varepsilon_{pqr}, \quad (4.29)$$

by definition of pseudotensor. All terms of the pqr sum will vanish except those where pqr is a permutation of 123, and when pqr is such a permutation the sum will correspond to the determinant of A except that its rows will have been permuted from 123 to ijk . This means that the pqr sum will have the value $\varepsilon_{ijk} \det(A)$, and

$$\eta'_{ijk} = \varepsilon_{ijk} [\det(A)]^2 = \varepsilon_{ijk}, \quad (4.30)$$

where the final result depends on the fact that $|\det(A)| = 1$. If the reader is uncomfortable with the above analysis, the result can be checked by enumeration of the contributions of the six permutations that correspond to nonzero values of η'_{ijk} .

Equation (4.30) not only shows that ε is a rank-3 pseudotensor, but that it is also isotropic. In other words, it has the same components in all rotated Cartesian coordinate systems, and -1 times those component values in all Cartesian systems that are reached by improper rotations. ■

Dual Tensors

With any **antisymmetric** second-rank tensor C (in 3-D space) we may associate a pseudovector $\mathbf{C}$ with components defined by

$$C_i = \frac{1}{2} \varepsilon_{ijk} C^{jk}. \quad (4.31)$$

In matrix form the antisymmetric $\mathbf{C}$ may be written

$$\mathbf{C} = \begin{pmatrix} 0 & C^{12} & -C^{31} \\ -C^{12} & 0 & C^{23} \\ C^{31} & -C^{23} & 0 \end{pmatrix}. \quad (4.32)$$

We know that C_i must transform as a vector under rotations because it was obtained from the double contraction of $\varepsilon_{ijk}C^{jk}$, but that it is really a pseudovector because of the pseudo nature of ε_{ijk} . Specifically, the components of $\mathbf{C}$ are given by

$$(C_1, C_2, C_3) = (C^{23}, C^{31}, C^{12}). \quad (4.33)$$

Note the cyclic order of the indices that comes from the cyclic order of the components of ε_{ijk} .

We identify the pseudovector of Eq. (4.33) and the antisymmetric tensor of Eq. (4.32) as **dual tensors**; they are simply different representations of the same information. Which of the dual pair we choose to use is a matter of convenience.

Here is another example of duality. If we take three vectors $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$, we may define the direct product

$$V^{ijk} = A^i B^j C^k. \quad (4.34)$$

V^{ijk} is evidently a rank-3 tensor. The dual quantity

$$V = \varepsilon_{ijk} V^{ijk} \quad (4.35)$$

is clearly a pseudoscalar. By expansion it is seen that

$$V = \begin{vmatrix} A^1 & B^1 & C^1 \\ A^2 & B^2 & C^2 \\ A^3 & B^3 & C^3 \end{vmatrix} \quad (4.36)$$

is our familiar scalar triple product.

Exercises

4.2.1 An antisymmetric square array is given by

$$\begin{pmatrix} 0 & C_3 & -C_2 \\ -C_3 & 0 & C_1 \\ C_2 & -C_1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & C^{12} & C^{13} \\ -C^{12} & 0 & C^{23} \\ -C^{13} & -C^{23} & 0 \end{pmatrix},$$

where (C_1, C_2, C_3) form a pseudovector. Assuming that the relation

$$C_i = \frac{1}{2!} \varepsilon_{ijk} C^{jk}$$

holds in all coordinate systems, prove that C^{jk} is a tensor. (This is another form of the quotient theorem.)

4.2.2 Show that the vector product is unique to 3-D space, that is, only in three dimensions can we establish a one-to-one correspondence between the components of an antisymmetric tensor (second-rank) and the components of a vector.

- 4.2.3** Write $\nabla \cdot \nabla \times \mathbf{A}$ and $\nabla \times \nabla \varphi$ in tensor (index) notation in $\mathbb{R}^3$ so that it becomes obvious that each expression vanishes.

$$\begin{aligned} \text{ANS. } \nabla \cdot \nabla \times \mathbf{A} &= \varepsilon_{ijk} \frac{\partial}{\partial x^i} \frac{\partial}{\partial x^j} A^k \\ (\nabla \times \nabla \varphi)_i &= \varepsilon_{ijk} \frac{\partial}{\partial x^j} \frac{\partial}{\partial x^k} \varphi. \end{aligned}$$

- 4.2.4** Verify that each of the following fourth-rank tensors is isotropic, that is, that it has the same form independent of any rotation of the coordinate systems.

- (a) $A_{jl}^{ik} = \delta_j^i \delta_l^k$,
- (b) $B_{kl}^{ij} = \delta_k^i \delta_l^j + \delta_l^i \delta_k^j$,
- (c) $C_{kl}^{ij} = \delta_k^i \delta_l^j - \delta_l^i \delta_k^j$.

- 4.2.5** Show that the two-index Levi-Civita symbol ε_{ij} is a second-rank pseudotensor (in two-dimensional [2-D] space). Does this contradict the uniqueness of δ_j^i (Exercise 4.1.4)?

- 4.2.6** Represent ε_{ij} by a 2×2 matrix, and using the 2×2 rotation matrix of Eq. (3.23), show that ε_{ij} is invariant under orthogonal similarity transformations.

- 4.2.7** Given $A_k = \frac{1}{2} \varepsilon_{ijk} B^{ij}$ with $B^{ij} = -B^{ji}$, antisymmetric, show that

$$B^{mn} = \varepsilon^{mnk} A_k.$$

4.3 TENSORS IN GENERAL COORDINATES

Metric Tensor

The distinction between contravariant and covariant transformations was established in Section 4.1, where we also observed that it only became meaningful when working with coordinate systems that are not Cartesian. We now want to examine relationships that can systematize the use of more general **metric spaces** (also called **Riemannian spaces**). Our initial illustrations will be for spaces with three dimensions.

Letting q^i denote coordinates in a general coordinate system, writing the index as a superscript to reflect the fact that coordinates transform contravariantly, we define **covariant basis vectors** $\mathbf{e}_i$ that describe the displacement (in Euclidean space) per unit change in q^i , keeping the other q^j constant. For the situations of interest here, both the direction and magnitude of $\mathbf{e}_i$ may be functions of position, so it is defined as the derivative

$$\mathbf{e}_i = \frac{\partial x}{\partial q^i} \hat{\mathbf{e}}_x + \frac{\partial y}{\partial q^i} \hat{\mathbf{e}}_y + \frac{\partial z}{\partial q^i} \hat{\mathbf{e}}_z. \quad (4.37)$$

An arbitrary vector $\mathbf{A}$ can now be formed as a linear combination of the basis vectors, multiplied by coefficients:

$$\mathbf{A} = A^1 \mathbf{e}_1 + A^2 \mathbf{e}_2 + A^3 \mathbf{e}_3. \quad (4.38)$$

At this point we have a linguistic ambiguity: $\mathbf{A}$ is a fixed object (usually called a vector) that may be described in various coordinate systems. But it is also customary to call the collection of coefficients A^i a vector (more specifically, a **contravariant vector**), while we have already called $\mathbf{e}_i$ a covariant basis vector. The important thing to observe here is that $\mathbf{A}$ is a fixed object that is not changed by our transformations, while its representation (the A^i) and the basis used for the representation (the $\mathbf{e}_i$) change in mutually inverse ways (as the coordinate system is changed) so as to keep $\mathbf{A}$ fixed.

Given our basis vectors, we can compute the displacement (change in position) associated with changes in the q^i . Because the basis vectors depend on position, our computation needs to be for small (infinitesimal) displacements ds . We have

$$(ds)^2 = \sum_{ij} (\mathbf{e}_i dq^i) \cdot (\mathbf{e}_j dq^j),$$

which, using the summation convention, can be written

$$(ds)^2 = g_{ij} dq^i dq^j, \quad (4.39)$$

with

$$g_{ij} = \mathbf{e}_i \cdot \mathbf{e}_j. \quad (4.40)$$

Since $(ds)^2$ is an invariant under rotational (and reflection) transformations, it is a scalar, and the quotient rule permits us to identify g_{ij} as a covariant tensor. Because of its role in defining displacement, g_{ij} is called the **covariant metric tensor**.

Note that the basis vectors can be defined by their Cartesian components, but they are, in general, neither unit vectors nor mutually orthogonal. Because they are often **not** unit vectors we have identified them by the symbol $\mathbf{e}$, not $\hat{\mathbf{e}}$. The lack of both a normalization and an orthogonality requirement means that g_{ij} , though manifestly symmetric, is not required to be diagonal, and its elements (including those on the diagonal) may be of either sign.

It is convenient to define a **contravariant** metric tensor that satisfies

$$g^{ik} g_{kj} = g_{jk} g^{ki} = \delta_j^i, \quad (4.41)$$

and is therefore the inverse of the covariant metric tensor. We will use g_{ij} and g^{ij} to make conversions between contravariant and covariant vectors that we then regard as related. Thus, we write

$$g_{ij} F^j = F_i \quad \text{and} \quad g^{ij} F_j = F^i. \quad (4.42)$$

Returning now to Eq. (4.38), we can manipulate it as follows:

$$\mathbf{A} = A^i \mathbf{e}_i = A^i \delta_i^k \mathbf{e}_k = (A^i g_{ij}) (g^{jk} \mathbf{e}_k) = A_j \mathbf{e}^j, \quad (4.43)$$

showing that the same vector can be represented either by contravariant or covariant components, with the two sets of components related by the transformation in Eq. (4.42).

Covariant and Contravariant Bases

We now define the **contravariant basis vectors**

$$\boldsymbol{\varepsilon}^i = \frac{\partial q^i}{\partial x} \hat{\mathbf{e}}_x + \frac{\partial q^i}{\partial y} \hat{\mathbf{e}}_y + \frac{\partial q^i}{\partial z} \hat{\mathbf{e}}_z, \quad (4.44)$$

giving them this name in anticipation of the fact that we can prove them to be the contravariant versions of the $\boldsymbol{\varepsilon}_i$. Our first step in this direction is to verify that

$$\boldsymbol{\varepsilon}^i \cdot \boldsymbol{\varepsilon}_j = \frac{\partial q^i}{\partial x} \frac{\partial x}{\partial q^j} + \frac{\partial q^i}{\partial y} \frac{\partial y}{\partial q^j} + \frac{\partial q^i}{\partial z} \frac{\partial z}{\partial q^j} = \delta_j^i, \quad (4.45)$$

a consequence of the chain rule and the fact that q^i and q^j are independent variables.

We next note that

$$(\boldsymbol{\varepsilon}^i \cdot \boldsymbol{\varepsilon}^j)(\boldsymbol{\varepsilon}_j \cdot \boldsymbol{\varepsilon}_k) = \delta_k^i, \quad (4.46)$$

also proved using the chain rule; the terms can be collected so that groups of them correspond to the identities in Eq. (4.45). Equation (4.46) shows that

$$g^{ij} = \boldsymbol{\varepsilon}^i \cdot \boldsymbol{\varepsilon}^j. \quad (4.47)$$

Multiplying both sides of Eq. (4.47) on the right by $\boldsymbol{\varepsilon}_j$ and performing the implied summation, the left-hand side of that equation, $g^{ij} \boldsymbol{\varepsilon}_j$, becomes the formula for $\boldsymbol{\varepsilon}^i$, while the right-hand side simplifies to the expression in Eq. (4.44), thereby proving that the contravariant vector in that equation was appropriately named.

We illustrate now some typical metric tensors and basis vectors in both covariant and contravariant form.

Example 4.3.1 SOME METRIC TENSORS

In spherical polar coordinates, $(q^1, q^2, q^3) \equiv (r, \theta, \varphi)$, and $x = r \sin \theta \cos \varphi$, $y = r \sin \theta \sin \varphi$, $z = r \cos \theta$. The covariant basis vectors are

$$\begin{aligned} \boldsymbol{\varepsilon}_r &= \sin \theta \cos \varphi \hat{\mathbf{e}}_x + \sin \theta \sin \varphi \hat{\mathbf{e}}_y + \cos \theta \hat{\mathbf{e}}_z, \\ \boldsymbol{\varepsilon}_\theta &= r \cos \theta \cos \varphi \hat{\mathbf{e}}_x + r \cos \theta \sin \varphi \hat{\mathbf{e}}_y - r \sin \theta \hat{\mathbf{e}}_z, \\ \boldsymbol{\varepsilon}_\varphi &= -r \sin \theta \sin \varphi \hat{\mathbf{e}}_x + r \sin \theta \cos \varphi \hat{\mathbf{e}}_y, \end{aligned}$$

and the contravariant basis vectors, which can be obtained in many ways, one of which is to start from $r^2 = x^2 + y^2 + z^2$, $\cos \theta = z/r$, $\tan \varphi = y/x$, are

$$\begin{aligned} \boldsymbol{\varepsilon}^r &= \sin \theta \cos \varphi \hat{\mathbf{e}}_x + \sin \theta \sin \varphi \hat{\mathbf{e}}_y + \cos \theta \hat{\mathbf{e}}_z, \\ \boldsymbol{\varepsilon}^\theta &= r^{-1} \cos \theta \cos \varphi \hat{\mathbf{e}}_x + r^{-1} \cos \theta \sin \varphi \hat{\mathbf{e}}_y - r^{-1} \sin \theta \hat{\mathbf{e}}_z, \\ \boldsymbol{\varepsilon}^\varphi &= -\frac{\sin \varphi}{r \sin \theta} \hat{\mathbf{e}}_x + \frac{\cos \varphi}{r \sin \theta} \hat{\mathbf{e}}_y, \end{aligned}$$

leading to

$$\begin{aligned} g_{11} &= \mathbf{e}_r \cdot \mathbf{e}_r = 1, \\ g_{22} &= \mathbf{e}_\theta \cdot \mathbf{e}_\theta = r^2, \\ g_{33} &= \mathbf{e}_\varphi \cdot \mathbf{e}_\varphi = r^2 \sin^2 \theta; \end{aligned}$$

all other g_{ij} vanish. Combining these to make g_{ij} and taking the inverse (to make g^{ij}), we have

$$(g_{ij}) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & r^2 & 0 \\ 0 & 0 & r^2 \sin^2 \theta \end{pmatrix}, \quad (g^{ij}) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & r^{-2} & 0 \\ 0 & 0 & (r \sin \theta)^{-2} \end{pmatrix}.$$

We can check that we have inverted g_{ij} correctly by comparing the expression given for g^{ij} from that built directly from $\mathbf{e}^i \cdot \mathbf{e}^j$. This check is left for the reader.

The Minkowski metric of special relativity has the form

$$(g_{ij}) = (g^{ij}) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}.$$

The motivation for including it in this example is to emphasize that for some metrics important in physics, distances ds^2 need not be positive (meaning that ds can be imaginary). ■

The relation between the covariant and contravariant basis vectors is useful for writing relationships between vectors. Let $\mathbf{A}$ and $\mathbf{B}$ be vectors with contravariant representations (A^i) and (B^i) . We may convert the representation of $\mathbf{B}$ to $B_i = g_{ij} B^j$, after which the scalar product $\mathbf{A} \cdot \mathbf{B}$ takes the form

$$\mathbf{A} \cdot \mathbf{B} = (A^i \mathbf{e}_i) \cdot (B_j \mathbf{e}^j) = A^i B_j (\mathbf{e}_i \cdot \mathbf{e}^j) = A^i B_i. \quad (4.48)$$

Another application is in writing the gradient in general coordinates. If a function ψ is given in a general coordinate system (q^i) , its gradient $\nabla \psi$ is a vector with Cartesian components

$$(\nabla \psi)_j = \frac{\partial \psi}{\partial q^i} \frac{\partial q^i}{\partial x^j}. \quad (4.49)$$

In vector notation, Eq. (4.49) becomes

$$\nabla \psi = \frac{\partial \psi}{\partial q^i} \mathbf{e}^i, \quad (4.50)$$

showing that the covariant representation of $\nabla \psi$ is the set of derivatives $\partial \psi / \partial q^i$. If we have reason to use a contravariant representation of the gradient, we can convert its components using Eq. (4.42).

Covariant Derivatives

Moving on to the derivatives of a vector, we find that the situation is much more complicated because the basis vectors $\mathbf{e}_i$ are in general not constant, and the derivative will not be a tensor whose components are the derivatives of the vector components.

Starting from the transformation rule for a contravariant vector,

$$(V')^i = \frac{\partial x^i}{\partial q^k} V^k,$$

and differentiating with respect to q^j , we get (for each i)

$$\frac{\partial (V')^i}{\partial q^j} = \frac{\partial x^i}{\partial q^k} \frac{\partial V^k}{\partial q^j} + \frac{\partial^2 x^i}{\partial q^j \partial q^k} V^k, \quad (4.51)$$

which appears to differ from the transformation law for a second-rank tensor because it contains a second derivative.

To see what to do next, let's write Eq. (4.51) as a single vector equation in the x_i coordinates, which we take to be Cartesian. The result is

$$\frac{\partial \mathbf{V}'}{\partial q^j} = \frac{\partial V^k}{\partial q^j} \mathbf{e}_k + V^k \frac{\partial \mathbf{e}_k}{\partial q^j}. \quad (4.52)$$

We now recognize that $\partial \mathbf{e}_k / \partial q^j$ must be some vector in the space spanned by the set of all $\mathbf{e}_i$ and we therefore write

$$\frac{\partial \mathbf{e}_k}{\partial q^j} = \Gamma_{jk}^\mu \mathbf{e}_\mu. \quad (4.53)$$

The quantities Γ_{jk}^μ are known as **Christoffel symbols of the second kind** (those of the first kind will be encountered shortly). Using the orthogonality property of the $\mathbf{e}$, Eq. (4.45), we can solve Eq. (4.53) by taking its dot product with any $\mathbf{e}^m$, reaching

$$\Gamma_{jk}^m = \mathbf{e}^m \cdot \frac{\partial \mathbf{e}_k}{\partial q^j}. \quad (4.54)$$

Moreover, we note that $\Gamma_{kj}^m = \Gamma_{jk}^m$, which can be demonstrated by writing out the components of $\partial \mathbf{e}_k / \partial q^j$.

Returning now to Eq. (4.52) and inserting Eq. (4.53), we initially get

$$\frac{\partial \mathbf{V}'}{\partial q^j} = \frac{\partial V^k}{\partial q^j} \mathbf{e}_k + V^k \Gamma_{jk}^\mu \mathbf{e}_\mu. \quad (4.55)$$

Interchanging the dummy indices k and μ in the last term of Eq. (4.55), we get the final result

$$\frac{\partial \mathbf{V}'}{\partial q^j} = \left(\frac{\partial V^k}{\partial q^j} + V^\mu \Gamma_{j\mu}^k \right) \mathbf{e}_k. \quad (4.56)$$

The parenthesized quantity in Eq. (4.56) is known as the **covariant derivative** of V , and it has (unfortunately) become standard to identify it by the awkward notation

$$V_{;j}^k = \frac{\partial V^k}{\partial q^j} + V^\mu \Gamma_{j\mu}^k, \quad \text{so} \quad \frac{\partial \mathbf{V}'}{\partial q^j} = V_{;j}^k \mathbf{e}_k. \quad (4.57)$$

If we rewrite Eq. (4.56) in the form

$$d\mathbf{V}' = \left[V_{;j}^k dq^j \right] \mathbf{e}_k,$$

and take note that dq^j is a contravariant vector, while $\mathbf{e}_k$ is covariant, we see that the covariant derivative, $V_{;j}^k$ is a mixed second-rank tensor.⁴ However, it is important to realize that although they bristle with indices, **neither** $\partial V^k / \partial q^j$ **nor** Γ_{jv}^k have individually the correct transformation properties to be tensors. It is only the combination in Eq. (4.57) that has the requisite transformational attributes.

It can be shown (see Exercise 4.3.6) that the covariant derivative of a **covariant** vector V_i is given by

$$V_{i;j} = \frac{\partial V_i}{\partial q^j} - V_k \Gamma_{ij}^k. \quad (4.58)$$

Like $V_{;j}^i$, $V_{i;j}$ is a second-rank tensor.

The physical importance of the covariant derivative is that it includes the changes in the basis vectors pursuant to a general dq^i , and is therefore more appropriate for describing physical phenomena than a formulation that considers only the changes in the coefficients multiplying the basis vectors.

Evaluating Christoffel Symbols

It may be more convenient to evaluate the Christoffel symbols by relating them to the metric tensor than simply to use Eq. (4.54). As an initial step in this direction, we define the Christoffel symbol of the **first kind** $[ij, k]$ by

$$[ij, k] \equiv g_{mk} \Gamma_{ij}^m, \quad (4.59)$$

from which the symmetry $[ij, k] = [ji, k]$ follows. Again, this $[ij, k]$ is not a third-rank tensor. Inserting Eq. (4.54) and applying the index-lowering transformation, Eq. (4.42), we have

$$\begin{aligned} [ij, k] &= g_{mk} \mathbf{e}^m \cdot \frac{\partial \mathbf{e}_i}{\partial q^j} \\ &= \mathbf{e}_k \cdot \frac{\partial \mathbf{e}_i}{\partial q^j}. \end{aligned} \quad (4.60)$$

Next, we write $g_{ij} = \mathbf{e}_i \cdot \mathbf{e}_j$ as in Eq. (4.40) and differentiate it, identifying the result with the aid of Eq. (4.60):

$$\begin{aligned} \frac{\partial g_{ij}}{\partial q^k} &= \frac{\partial \mathbf{e}_i}{\partial q^k} \cdot \mathbf{e}_j + \mathbf{e}_i \cdot \frac{\partial \mathbf{e}_j}{\partial q^k} \\ &= [ik, j] + [jk, i]. \end{aligned}$$

⁴ $\mathbf{V}'$ does not contribute to the covariant/contravariant character of the equation as its implicit index labels the Cartesian coordinates, as is also the case for $\mathbf{e}_k$.

We then note that we can combine three of these derivatives with different index sets, with a result that simplifies to give

$$\frac{1}{2} \left[\frac{\partial g_{ik}}{\partial q^j} + \frac{\partial g_{jk}}{\partial q^i} - \frac{\partial g_{ij}}{\partial q^k} \right] = [ij, k]. \quad (4.61)$$

We now return to Eq. (4.59), which we solve for Γ_{ij}^m by multiplying both sides by g^{nk} , summing over k , and using the fact that $(g_{\mu\nu})$ and $(g^{\mu\nu})$ are mutually inverse, see Eq. (4.41):

$$\Gamma_{ij}^n = \sum_k g^{nk} [ij, k]. \quad (4.62)$$

Finally, substituting for $[ij, k]$ from Eq. (4.61), and once again using the summation convention, we get:

$$\Gamma_{ij}^n = g^{nk} [ij, k] = \frac{1}{2} g^{nk} \left[\frac{\partial g_{ik}}{\partial q^j} + \frac{\partial g_{jk}}{\partial q^i} - \frac{\partial g_{ij}}{\partial q^k} \right]. \quad (4.63)$$

The apparatus of this subsection becomes unnecessary in Cartesian coordinates, because the basis vectors have vanishing derivatives, and the covariant and ordinary partial derivatives then coincide.

Tensor Derivative Operators

With covariant differentiation now available, we are ready to derive the vector differential operators in general tensor form.

Gradient—We have already discussed it, with the result from Eq. (4.50):

$$\nabla \psi = \frac{\partial \psi}{\partial q^i} \mathbf{e}^i. \quad (4.64)$$

Divergence—A vector $\mathbf{V}$ whose contravariant representation is $V^i \mathbf{e}_i$ has divergence

$$\nabla \cdot \mathbf{V} = \mathbf{e}^j \cdot \frac{\partial (V^i \mathbf{e}_i)}{\partial q^j} = \mathbf{e}^j \cdot \left(\frac{\partial V^i}{\partial q^j} + V^k \Gamma_{jk}^i \right) \mathbf{e}_i = \frac{\partial V^i}{\partial q^i} + V^k \Gamma_{ik}^i. \quad (4.65)$$

Note that the covariant derivative has appeared here. Expressing Γ_{ik}^i by Eq. (4.63), we have

$$\Gamma_{ik}^i = \frac{1}{2} g^{im} \left[\frac{\partial g_{im}}{\partial q^k} + \frac{\partial g_{km}}{\partial q^i} - \frac{\partial g_{ik}}{\partial q^m} \right] = \frac{1}{2} g^{im} \frac{\partial g_{im}}{\partial q^k}, \quad (4.66)$$

where we have recognized that the last two terms in the bracket will cancel because by changing the names of their dummy indices they can be identified as identical except in sign.

Because (g^{im}) is the matrix inverse to (g_{im}) , we note that the combination of matrix elements on the right-hand side of Eq. (4.66) is similar to those in the formula for the derivative of a determinant, Eq. (2.35); remember that g is symmetric: $g^{im} = g^{mi}$. In the present notation, the relevant formula is

$$\frac{d \det(g)}{dq^k} = \det(g) g^{im} \frac{\partial g_{im}}{\partial q^k}, \quad (4.67)$$

where $\det(g)$ is the determinant of the **covariant** metric tensor $(g_{\mu\nu})$. Using Eq. (4.67), Eq. (4.66) becomes

$$\Gamma_{ik}^i = \frac{1}{2\det(g)} \frac{d\det(g)}{dq^k} = \frac{1}{[\det(g)]^{1/2}} \frac{\partial[\det(g)]^{1/2}}{\partial q^k}. \quad (4.68)$$

Combining the result in Eq. (4.68) with Eq. (4.65), we obtain a maximally compact formula for the divergence of a contravariant vector $\mathbf{V}$:

$$\nabla \cdot \mathbf{V} = V_{;i}^i = \frac{1}{[\det(g)]^{1/2}} \frac{\partial}{\partial q^k} \left([\det(g)]^{1/2} V^k \right). \quad (4.69)$$

To compare this result with that for an orthogonal coordinate system, Eq. (3.141), note that $\det(g) = (h_1 h_2 h_3)^2$ and that the k component of the vector represented by $\mathbf{V}$ in Eq. (3.141) is, in the present notation, equal to $V^k |\mathbf{e}_k| = h_k V^k$ (no summation).

Laplacian—We can form the Laplacian $\nabla^2 \psi$ by inserting an expression for the gradient $\nabla \psi$ into the formula for the divergence, Eq. (4.69). However, that equation uses the contravariant coefficients V^k , so we must describe the gradient in its contravariant representation. Since Eq. (4.64) shows that the covariant coefficients of the gradient are the derivatives $\partial \psi / \partial q^i$, its contravariant coefficients have to be

$$g^{ki} \frac{\partial \psi}{\partial q^i}.$$

Insertion into Eq. (4.69) then yields

$$\nabla^2 \psi = \frac{1}{[\det(g)]^{1/2}} \frac{\partial}{\partial q^k} \left([\det(g)]^{1/2} g^{ki} \frac{\partial \psi}{\partial q^i} \right). \quad (4.70)$$

For **orthogonal** systems the metric tensor is diagonal and the contravariant $g^{ii} = (h_i)^{-2}$ (no summation). Equation (4.70) then reduces to

$$\nabla \cdot \nabla \psi = \frac{1}{h_1 h_2 h_3} \frac{\partial}{\partial q^i} \left(\frac{h_1 h_2 h_3}{h_i^2} \frac{\partial \psi}{\partial q^i} \right),$$

in agreement with Eq. (3.142).

Curl—The difference of derivatives that appears in the curl has components that can be written

$$\frac{\partial V_i}{\partial q^j} - \frac{\partial V_j}{\partial q^i} = \frac{\partial V_i}{\partial q^j} - V_k \Gamma_{ij}^k - \frac{\partial V_j}{\partial q^i} + V_k \Gamma_{ji}^k = V_{i;j} - V_{j;i}, \quad (4.71)$$

where we used the symmetry of the Christoffel symbols to obtain a cancellation. The reason for the manipulation in Eq. (4.71) is to bring all the terms on its right-hand side to tensor form. In using Eq. (4.71), it is necessary to remember that the quantities V_i are coefficients of the possibly nonunit $\mathbf{e}^i$ and are therefore **not** components of $\mathbf{V}$ in the orthonormal basis $\hat{\mathbf{e}}_i$.

Exercises

- 4.3.1** For the special case of 3-D space ($\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$ defining a right-handed coordinate system, not necessarily orthogonal), show that

$$\mathbf{e}^i = \frac{\mathbf{e}_j \times \mathbf{e}_k}{\mathbf{e}_j \times \mathbf{e}_k \cdot \mathbf{e}_i}, \quad i, j, k = 1, 2, 3 \text{ and cyclic permutations.}$$

Note. These contravariant basis vectors $\mathbf{e}^i$ define the reciprocal lattice space of Example 3.2.1.

- 4.3.2** If the covariant vectors $\mathbf{e}_i$ are orthogonal, show that

- (a) g_{ij} is diagonal,
- (b) $g^{ii} = 1/g_{ii}$ (no summation),
- (c) $|\mathbf{e}^i| = 1/|\mathbf{e}_i|$.

- 4.3.3** Prove that $(\mathbf{e}^i \cdot \mathbf{e}^j)(\mathbf{e}_j \cdot \mathbf{e}_k) = \delta_k^i$.

- 4.3.4** Show that $\Gamma_{jk}^m = \Gamma_{kj}^m$.

- 4.3.5** Derive the covariant and contravariant metric tensors for circular cylindrical coordinates.

- 4.3.6** Show that the covariant derivative of a covariant vector is given by

$$V_{i;j} \equiv \frac{\partial V_i}{\partial q^j} - V_k \Gamma_{ij}^k.$$

Hint. Differentiate

$$\mathbf{e}^i \cdot \mathbf{e}_j = \delta_j^i.$$

- 4.3.7** Verify that $V_{i;j} = g_{ik} V_{;j}^k$ by showing that

$$\frac{\partial V_i}{\partial q^j} - V_k \Gamma_{ij}^k = g_{ik} \left[\frac{\partial V^k}{\partial q^j} + V^m \Gamma_{mj}^k \right].$$

- 4.3.8** From the circular cylindrical metric tensor g_{ij} , calculate the Γ_{ij}^k for circular cylindrical coordinates.

Note. There are only three nonvanishing Γ .

- 4.3.9** Using the Γ_{ij}^k from [Exercise 4.3.8](#), write out the covariant derivatives $V_{;j}^i$ of a vector $\mathbf{V}$ in circular cylindrical coordinates.

- 4.3.10** Show that for the metric tensor $g_{ij;k} = g_{;k}^{ij} = 0$.

- 4.3.11** Starting with the divergence in tensor notation, [Eq. \(4.70\)](#), develop the divergence of a vector in spherical polar coordinates, [Eq. \(3.157\)](#).

- 4.3.12** The covariant vector A_i is the gradient of a scalar. Show that the difference of covariant derivatives $A_{i;j} - A_{j;i}$ vanishes.

4.4 JACOBIANS

In the preceding chapters we have considered the use of curvilinear coordinates, but have not placed much focus on transformations between coordinate systems, and in particular on the way in which multidimensional integrals must transform when the coordinate system is changed. To provide formulas that will be useful in spaces with arbitrary numbers of dimensions, and with transformations involving coordinate systems that are not orthogonal, we now return to the notion of the **Jacobian**, introduced but not fully developed in Chapter 1.

As already mentioned in Chapter 1, changes of variables in multiple integrations, say from variables $x_1, x_2, \dots$ to $u_1, u_2, \dots$ requires that we replace the differential $dx_1 dx_2 \dots$ with $J du_1 du_2 \dots$, where J , called the Jacobian, is the quantity (usually dependent on the variables) needed to make these expressions mutually consistent. More specifically, we identify $d\tau = J du_1 du_2 \dots$ as the “volume” of a region of width du_1 in u_1 , du_2 in $u_2, \dots$, where the “volume” is to be computed in the $x_1, x_2, \dots$ space, treated as Cartesian coordinates.

To obtain a formula for J we start by identifying the displacement (in the Cartesian system defined by the x_i) that corresponds to a change in each variable u_i . Letting $d\mathbf{s}(u_i)$ be that displacement (which is a vector), we can decompose it into Cartesian components as follows:

$$\begin{aligned} d\mathbf{s}(u_1) &= \left[\left(\frac{\partial x_1}{\partial u_1} \right) \hat{\mathbf{e}}_1 + \left(\frac{\partial x_2}{\partial u_1} \right) \hat{\mathbf{e}}_2 + \dots \right] du_1, \\ d\mathbf{s}(u_2) &= \left[\left(\frac{\partial x_1}{\partial u_2} \right) \hat{\mathbf{e}}_1 + \left(\frac{\partial x_2}{\partial u_2} \right) \hat{\mathbf{e}}_2 + \dots \right] du_2, \\ d\mathbf{s}(u_3) &= \left[\left(\frac{\partial x_1}{\partial u_3} \right) \hat{\mathbf{e}}_1 + \left(\frac{\partial x_2}{\partial u_3} \right) \hat{\mathbf{e}}_2 + \dots \right] du_3, \\ \dots\dots &= \dots\dots\dots \end{aligned} \quad (4.72)$$

The partial derivatives $(\partial x_i / \partial u_j)$ in Eq. (4.72) must be understood to be evaluated with the other u_k held constant. It would clutter the formula an unreasonable amount to indicate this explicitly.

If we had only two variables, u_1 and u_2 , the differential area would simply be $|d\mathbf{s}(u_1)|$ times the component of $d\mathbf{s}(u_2)$ that is perpendicular to $d\mathbf{s}(u_1)$. If there were a third variable, u_3 , we would further multiply by the component of $d\mathbf{s}(u_3)$ that was perpendicular to both $d\mathbf{s}(u_1)$ and $d\mathbf{s}(u_2)$. Extension to arbitrary numbers of dimensions is obvious.

What is less obvious is an explicit formula for the “volume” for an arbitrary number of dimensions. Let’s start by writing Eq. (4.72) in matrix form:

$$\begin{pmatrix} \frac{d\mathbf{s}(u_1)}{du_1} \\ \frac{d\mathbf{s}(u_2)}{du_2} \\ \frac{d\mathbf{s}(u_3)}{du_3} \\ \dots \end{pmatrix} = \begin{pmatrix} \frac{\partial x_1}{\partial u_1} & \frac{\partial x_2}{\partial u_1} & \frac{\partial x_3}{\partial u_1} & \dots \\ \frac{\partial x_1}{\partial u_2} & \frac{\partial x_2}{\partial u_2} & \frac{\partial x_3}{\partial u_2} & \dots \\ \frac{\partial x_1}{\partial u_3} & \frac{\partial x_2}{\partial u_3} & \frac{\partial x_3}{\partial u_3} & \dots \\ \dots & \dots & \dots & \dots \end{pmatrix} \begin{pmatrix} \hat{\mathbf{e}}_1 \\ \hat{\mathbf{e}}_2 \\ \hat{\mathbf{e}}_3 \\ \dots \end{pmatrix}. \quad (4.73)$$

We now proceed to make changes to the second and succeeding rows of the square matrix in Eq. (4.73) that may destroy the relation to the $ds(u_i)/du_i$, but which will leave the “volume” unchanged. In particular, we subtract from the second row of the derivative matrix that multiple of the first row which will cause the first element of the modified second row to vanish. This will not change the “volume” because it modifies $ds(u_2)/du_2$ by adding or subtracting a vector in the $ds(u_1)/du_1$ direction, and therefore does not affect the component of $ds(u_2)/du_2$ perpendicular to $ds(u_1)/du_1$. See Fig. 4.1.

The alert reader will recall that this modification of the second row of our matrix is an operation that was used when evaluating determinants, and was there justified because it did not change the value of the determinant. We have a similar situation here; the operation will not change the value of the differential “volume” because we are changing only the component of $ds(u_2)/du_2$ that is in the $ds(u_1)/du_1$ direction. In a similar fashion, we can carry out further operations of the same kind that will lead to a matrix in which all the elements below the principal diagonal have been reduced to zero. The situation at this point is indicated schematically for an 4-D space as the transition from the first to the second matrix in Fig. 4.2. These modified $ds(u_i)/du_i$ will lead to the same differential volume as the original $ds(u_i)/du_i$. This modified matrix will no longer provide a faithful representation of the differential region in the u_i space, but that is irrelevant since our only objective is to evaluate the differential “volume.”

We next take the final (n th) row of our modified matrix, which will be entirely zero except for its last element, and subtract a suitable multiple of it from all the other rows to introduce zeros in the last element of every row above the principal diagonal. These operations correspond to changes in which we modify only the components of the other $ds(u_i)/du_i$ that are in the direction of $ds(u_n)$, and therefore will not change the differential “volume.” Then, using the next-to-last row (which now has only a diagonal element), we can in a similar fashion introduce zeros in the next-to-last column of all the preceding rows. Continuing this process, we will ultimately have a set of modified $ds(u_i)/du_i$ that will have the structure shown as the last matrix in Fig. 4.2. Because our modified matrix is diagonal, with each nonzero element associated with a single different $\hat{e}_i$, the “volume” is

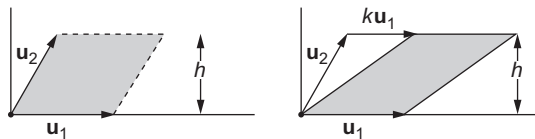


FIGURE 4.1 Area remains unchanged when vector proportional to $\mathbf{u}_1$ is added to $\mathbf{u}_2$.

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{pmatrix} \rightarrow \begin{pmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ 0 & b_{22} & b_{23} & b_{24} \\ 0 & 0 & b_{33} & b_{34} \\ 0 & 0 & 0 & b_{44} \end{pmatrix} \rightarrow \begin{pmatrix} a_{11} & 0 & 0 & 0 \\ 0 & b_{22} & 0 & 0 \\ 0 & 0 & b_{33} & 0 \\ 0 & 0 & 0 & b_{44} \end{pmatrix}$$

FIGURE 4.2 Manipulation of Jacobian matrix. Here $a_{ij} = (\partial x_j / \partial u_i)$, and b_{ij} are formed by combining rows (see text).

then easily computed as the product of the diagonal elements. This product of the diagonal elements of a diagonal matrix is an evaluation of its determinant.

Reviewing what we have done, we see that we have identified the differential “volume” as a quantity which is equal to the determinant of the original derivative set. This must be so, because we obtained our final result by carrying out operations each of which leaves a determinant unchanged. The final result can be expressed as the well-known formula for the Jacobian:

$$d\tau = J du_1 du_2 \dots, \quad J = \begin{vmatrix} \frac{\partial x_1}{\partial u_1} & \frac{\partial x_2}{\partial u_1} & \frac{\partial x_3}{\partial u_1} & \dots \\ \frac{\partial x_1}{\partial u_2} & \frac{\partial x_2}{\partial u_2} & \frac{\partial x_3}{\partial u_2} & \dots \\ \frac{\partial x_1}{\partial u_3} & \frac{\partial x_2}{\partial u_3} & \frac{\partial x_3}{\partial u_3} & \dots \\ \dots & \dots & \dots & \dots \end{vmatrix} \equiv \frac{\partial(x_1, x_2, \dots)}{\partial(u_1, u_2, \dots)}. \quad (4.74)$$

The standard notation for the Jacobian, shown as the last member of Eq. (4.74), is a convenient reminder of the way in which the partial derivatives appear in it. Note also that when the standard notation for J is inserted in the expression for $d\tau$, the overall expression has $du_1 du_2 \dots$ in the numerator, while $\partial(u_1, u_2, \dots)$ appears in the denominator. This feature can help the user to make a proper identification of the Jacobian.

A few words about nomenclature: The matrix in Eq. (4.73) is sometimes called the **Jacobian matrix**, with the determinant in Eq. (4.74) then distinguished by calling it the **Jacobian determinant**. Unless within a discussion in which both these quantities appear and need to be separately identified, most authors simply call J , the determinant in Eq. (4.74), **the Jacobian**. That is the usage we follow in this book.

We close with one final observation. Since J is a determinant, it will have a sign that depends on the order in which the x_i and u_i are specified. This ambiguity corresponds to our freedom to choose either right- or left-handed coordinates. In typical applications involving a Jacobian, it is usual to take its absolute value and to choose the ranges of the individual u_i integrals in a way that gives the correct sign for the overall integral.

Example 4.4.1 2-D and 3-D JACOBIANS

In two dimensions, with Cartesian coordinates x, y and transformed coordinates u, v , the element of area dA has, following Eq. (4.74), the form

$$dA = du dv \left[\left(\frac{\partial x}{\partial u} \right) \left(\frac{\partial y}{\partial v} \right) - \left(\frac{\partial x}{\partial v} \right) \left(\frac{\partial y}{\partial u} \right) \right].$$

This is the expected result, as the quantity in square brackets is the formula for the z component of the cross product of the two vectors

$$\left(\frac{\partial x}{\partial u} \right) \hat{\mathbf{e}}_x + \left(\frac{\partial y}{\partial u} \right) \hat{\mathbf{e}}_y \quad \text{and} \quad \left(\frac{\partial x}{\partial v} \right) \hat{\mathbf{e}}_x + \left(\frac{\partial y}{\partial v} \right) \hat{\mathbf{e}}_y,$$

and it is well known that the magnitude of the cross product of two vectors is a measure of the area of the parallelogram with sides formed by the vectors.

In three dimensions, the determinant in the Jacobian corresponds exactly with the formula for the scalar triple product, Eq. (3.12). Letting A_x, A_y, A_z in that formula refer to the derivatives $(\partial x/\partial u), (\partial y/\partial u), (\partial z/\partial u)$, with the components of $\mathbf{B}$ and $\mathbf{C}$ similarly related to derivatives with respect to v and w , we recover the formula for the volume within the parallelepiped defined by three vectors. ■

Inverse of Jacobian

Since the x_i and the u_i are arbitrary sets of coordinates, we could have carried out the entire analysis of the preceding subsection regarding the u_i as the fundamental coordinate system, with the x_i as coordinates reached by a change of variables. In that case, our Jacobian (which we choose to label J^{-1}), would be

$$J^{-1} = \frac{\partial(u_1, u_2, \dots)}{\partial(x_1, x_2, \dots)}. \quad (4.75)$$

It is clear that if $dx_1 dx_2 \dots = J du_1 du_2 \dots$, then it must also be true that $du_1 du_2 \dots = (1/J) dx_1 dx_2 \dots$. Let's verify that the quantity we have called J^{-1} is in fact $1/J$.

Let's represent the two Jacobian **matrices** involved here as

$$A = \begin{pmatrix} \frac{\partial x_1}{\partial u_1} & \frac{\partial x_2}{\partial u_1} & \frac{\partial x_3}{\partial u_1} & \dots \\ \frac{\partial x_1}{\partial u_2} & \frac{\partial x_2}{\partial u_2} & \frac{\partial x_3}{\partial u_2} & \dots \\ \frac{\partial x_1}{\partial u_3} & \frac{\partial x_2}{\partial u_3} & \frac{\partial x_3}{\partial u_3} & \dots \\ \dots & \dots & \dots & \dots \end{pmatrix}, \quad B = \begin{pmatrix} \frac{\partial u_1}{\partial x_1} & \frac{\partial u_2}{\partial x_1} & \frac{\partial u_3}{\partial x_1} & \dots \\ \frac{\partial u_1}{\partial x_2} & \frac{\partial u_2}{\partial x_2} & \frac{\partial u_3}{\partial x_2} & \dots \\ \frac{\partial u_1}{\partial x_3} & \frac{\partial u_2}{\partial x_3} & \frac{\partial u_3}{\partial x_3} & \dots \\ \dots & \dots & \dots & \dots \end{pmatrix}.$$

We then have $J = \det(A)$ and $J^{-1} = \det(B)$. We would like to show that $J J^{-1} = \det(A) \det(B) = 1$. The proof is fairly simple if we use the determinant product theorem. Thus, we write

$$\det(A) \det(B) = \det(AB),$$

and now all we need show is that the matrix product AB is a unit matrix. Carrying out the matrix multiplication, we find, as a result of the chain rule,

$$(AB)_{ij} = \sum_k \left(\frac{\partial x_k}{\partial u_i} \right) \left(\frac{\partial u_j}{\partial x_k} \right) = \left(\frac{\partial u_j}{\partial u_i} \right) = \delta_{ij}, \quad (4.76)$$

verifying that AB is indeed a unit matrix.

The relation between the Jacobian and its inverse is of practical interest. It may turn out that the derivatives $\partial u_i / \partial x_j$ are easier to compute than $\partial x_i / \partial u_j$, making it convenient to obtain J by first constructing and evaluating the determinant for J^{-1} .

Example 4.4.2 DIRECT AND INVERSE APPROACHES TO JACOBIAN

Suppose we need the Jacobian $\frac{\partial(r, \theta, \varphi)}{\partial(x, y, z)}$, where x, y , and z are Cartesian coordinates and r, θ, φ are spherical polar coordinates. Using Eq. (4.74) and the relations

$$r = \sqrt{x^2 + y^2 + z^2}, \quad \theta = \cos^{-1} \left(\frac{z}{\sqrt{x^2 + y^2 + z^2}} \right), \quad \varphi = \tan^{-1} \left(\frac{y}{x} \right),$$

we find after significant effort (letting $\rho^2 = x^2 + y^2$),

$$J = \frac{\partial(r, \theta, \varphi)}{\partial(x, y, z)} = \begin{vmatrix} \frac{x}{r} & \frac{y}{r} & \frac{z}{r} \\ \frac{xz}{r^2 \rho} & \frac{yz}{r^2 \rho} & -\frac{\rho}{r^2} \\ -\frac{y}{\rho^2} & \frac{x}{\rho^2} & 0 \end{vmatrix} = \frac{1}{r\rho} = \frac{1}{r^2 \sin \theta}.$$

It is much less effort to use the relations

$$x = r \sin \theta \cos \varphi, \quad y = r \sin \theta \sin \varphi, \quad z = r \cos \theta,$$

and then to evaluate (easily),

$$J^{-1} = \frac{\partial(x, y, z)}{\partial(r, \theta, \varphi)} = \begin{vmatrix} \sin \theta \cos \varphi & \sin \theta \sin \varphi & \cos \theta \\ r \cos \theta \cos \varphi & r \cos \theta \sin \varphi & -r \sin \theta \\ -r \sin \theta \sin \varphi & r \sin \theta \cos \varphi & 0 \end{vmatrix} = r^2 \sin \theta.$$

We finish by writing $J = 1/J^{-1} = 1/r^2 \sin \theta$. ■

Exercises

4.4.1 Assuming the functions u and v to be differentiable,

- (a) Show that a necessary and sufficient condition that $u(x, y, z)$ and $v(x, y, z)$ are related by some function $f(u, v) = 0$ is that $(\nabla u) \times (\nabla v) = 0$;
- (b) If $u = u(x, y)$ and $v = v(x, y)$, show that the condition $(\nabla u) \times (\nabla v) = 0$ leads to the 2-D Jacobian

$$J = \frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix} = 0.$$

4.4.2 A 2-D orthogonal system is described by the coordinates q_1 and q_2 . Show that the Jacobian J satisfies the equation

$$J \equiv \frac{\partial(x, y)}{\partial(q_1, q_2)} \equiv \frac{\partial x}{\partial q_1} \frac{\partial y}{\partial q_2} - \frac{\partial x}{\partial q_2} \frac{\partial y}{\partial q_1} = h_1 h_2.$$

Hint. It's easier to work with the square of each side of this equation.

- 4.4.3** For the transformation $u = x + y$, $v = x/y$, with $x \geq 0$ and $y \geq 0$, find the Jacobian $\frac{\partial(x, y)}{\partial(u, v)}$
- (a) By direct computation,
 - (b) By first computing J^{-1} .

4.5 DIFFERENTIAL FORMS

Our study of tensors has indicated that significant complications arise when we leave Cartesian coordinate systems, even in traditional contexts such as the introduction of spherical or cylindrical coordinates. Much of the difficulty arises from the fact that the metric (as expressed in a coordinate system) becomes position-dependent, and that the lines or surfaces of constant coordinate values become curved. Many of the most vexing problems can be avoided if we work in a geometry that deals with infinitesimal displacements, because the situations of most importance in physics then become locally similar to the simpler and more familiar conditions based on Cartesian coordinates.

The calculus of differential forms, of which the leading developer was Elie Cartan, has become recognized as a natural and very powerful tool for the treatment of curved coordinates, both in classical settings and in contemporary studies of curved space-time. Cartan's calculus leads to a remarkable unification of concepts and theorems of vector analysis that is worth pursuing, with the result that in differential geometry and in theoretical physics the use of differential forms is now widespread.

Differential forms provide an important entry to the role of geometry in physics, and the connectivity of the spaces under discussion (technically, referred to as their **topology**) has physical implications. Illustrations are provided already by situations as simple as the fact that a coordinate defined on a circle cannot be single-valued and continuous at all angles. More sophisticated consequences of topology in physics, largely beyond the scope of the present text, include gauge transformations, flux quantization, the Bohm-Aharonov effect, emerging theories of elementary particles, and phenomena of general relativity.

Introduction

For simplicity we begin our discussion of differential forms in a notation appropriate for ordinary 3-D space, though the real power of the methods under study is that they are not limited either by the dimensionality of the space or by its metric properties (and are therefore also relevant to the curved space-time of general relativity). The basic quantities under consideration are the **differentials** dx , dy , dz (identified with linearly independent directions in the space), linear combinations thereof, and more complicated quantities built from these by combination rules we will shortly discuss in detail. Taking for example dx , it is essential to understand that in our current context it is not just an infinitesimal number describing a change in the x coordinate, but is to be viewed as a mathematical object with certain operational properties (which, admittedly, may include its eventual use in contexts such as the evaluation of line, surface, or volume integrals). The rules by which dx and

related quantities can be manipulated have been designed to permit expressions such as

$$\omega = A(x, y, z) dx + B(x, y, z) dy + C(x, y, z) dz, \quad (4.77)$$

which are called **1-forms**, to be related to quantities that occur as the integrands of line integrals, to permit expressions of the type

$$\omega = F(x, y, z) dx \wedge dy + G(x, y, z) dx \wedge dz + H(x, y, z) dy \wedge dz, \quad (4.78)$$

which are called **2-forms**, to be related to the integrands of surface integrals, and to permit expressions like

$$\omega = K(x, y, z) dx \wedge dy \wedge dz, \quad (4.79)$$

known as **3-forms**, to be related to the integrands of volume integrals.

The $\wedge$ symbol (called “wedge”) indicates that the individual differentials are to be combined to form more complicated objects using the rules of **exterior algebra** (sometimes called **Grassmann algebra**), so more is being implied by Eqs. (4.77) to (4.79) than the somewhat similar formulas that might appear in the conventional notation for various kinds of integrals. To maintain contact with other presentations on differential forms, we note that some authors omit the wedge symbol, thereby assuming that the reader knows that the differentials are to be combined according to the rules of exterior algebra. In order to minimize potential confusion, we will continue to write the wedge symbol for these combinations of differentials (which are called **exterior**, or **wedge** products).

To write differential forms in ways that do not presuppose the dimension of the underlying space, we sometimes write the differentials as dx_i , designating a form as a **p -form** if it contains p factors dx_i . Ordinary functions (containing no dx_i) can be identified as **0-forms**.

The mathematics of differential forms was developed with the aim of systematizing the application of calculus to **differentiable manifolds**, loosely defined as sets of points that can be identified by coordinates that locally vary “smoothly” (meaning that they are differentiable to whatever degree is needed for analysis).⁵ We are presently focusing attention on the differentials that appear in the forms; one could also consider the behavior of the coefficients. For example, when we write the 1-form

$$\omega = A_x dx + A_y dy + A_z dz,$$

A_x, A_y, A_z will behave under a coordinate transformation like the components of a vector, and in the older differential-forms literature the differentials and the coefficients were referred to as contravariant and covariant vector components, since these two sets of quantities must transform in mutually inverse ways under rotations of the coordinate system. What is relevant for us at this point is that relationships we develop for differential forms can be translated into related relationships for their vector coefficients, yielding not only various well-known formulas of vector analysis but also showing how they can be generalized to spaces of higher dimension.

⁵A manifold defined on a circle or sphere must have a coordinate that cannot be globally smooth (in the usual coordinate systems it will jump somewhere by 2π). This and related issues connect topology and physics, and are for the most part outside the scope of this text.

Exterior Algebra

The central idea in exterior algebra is that the operations are designed to create permutational antisymmetry. Assuming the basis 1-forms are dx_i , that ω_j are arbitrary p -forms (of respective orders p_j), and that a and b are ordinary numbers or functions, the wedge product is defined to have the properties

$$\begin{aligned} (a\omega_1 + b\omega_2) \wedge \omega_3 &= a\omega_1 \wedge \omega_3 + b\omega_2 \wedge \omega_3 \quad (p_1 = p_2), \\ (\omega_1 \wedge \omega_2) \wedge \omega_3 &= \omega_1 \wedge (\omega_2 \wedge \omega_3), \quad a(\omega_1 \wedge \omega_2) = (a\omega_1) \wedge \omega_2, \\ dx_i \wedge dx_j &= -dx_j \wedge dx_i. \end{aligned} \tag{4.80}$$

We thus have the usual associative and distributive laws, and each term of an arbitrary differential form can be reduced to a coefficient multiplying a dx_i or a wedge product of the generic form

$$dx_i \wedge dx_j \wedge \cdots \wedge dx_p.$$

Moreover, the properties in Eq. (4.80) permit all the coefficient functions to be collected at the beginning of a form. For example,

$$a dx_1 \wedge b dx_2 = -a(b dx_2 \wedge dx_1) = -ab(dx_2 \wedge dx_1) = ab(dx_1 \wedge dx_2).$$

We therefore generally do not need parentheses to indicate the order in which products are to be carried out.

We can use the last of Eqs. (4.80) to bring the index set into any desired order. If any two of the dx_i are the same, the expression will vanish because $dx_i \wedge dx_i = -dx_i \wedge dx_i = 0$; otherwise, the ordered-index form will have a sign determined by the parity of the index permutation needed to obtain the ordering. It is **not** a coincidence that this is the sign rule for the terms of a determinant, compare Eq. (2.10). Letting ε_P stand for the Levi-Civita symbol for the permutation to ascending index order, an arbitrary wedge product of dx_i can, for example, be brought to the form

$$\varepsilon_P dx_{h_1} \wedge dx_{h_2} \wedge \cdots \wedge dx_{h_p}, \quad 1 \leq h_1 < h_2 < \cdots < h_p.$$

If any of the dx_i in a differential form is linearly dependent on the others, then its expansion into linearly independent terms will produce a duplicated dx_j and cause the form to vanish. Since the number of linearly independent dx_j cannot be larger than the dimension of the underlying space, we see that in a space of dimension d we only need to consider p -forms with $p \leq d$. Thus, in 3-D space, only up through 3-forms are relevant; for Minkowski space (ct, x, y, z) , we will also have 4-forms.

Example 4.5.1 SIMPLIFYING DIFFERENTIAL FORMS

Consider the wedge product

$$\begin{aligned} \omega &= (3dx + 4dy - dz) \wedge (dx - dy + 2dz) = 3dx \wedge dx - 3dx \wedge dy + 6dx \wedge dz \\ &\quad + 4dy \wedge dx - 4dy \wedge dy + 8dy \wedge dz - dz \wedge dx + dz \wedge dy - 2dz \wedge dz. \end{aligned}$$

The terms with duplicate differentials, e.g., $dx \wedge dx$, vanish, and products that differ only in the order of the 1-forms can be combined, changing the sign of the product when we interchange its factors. We get

$$\omega = -7dx \wedge dy + 7dx \wedge dz + 7dy \wedge dz = 7(dy \wedge dz - dz \wedge dx - dx \wedge dy).$$

We will shortly see that in three dimensions there are some advantages to bringing the 1-forms into cyclic order (rather than ascending or descending order) in the wedge products, and we did so in the final simplification of ω . ■

The antisymmetry built into the exterior algebra has an important purpose: It causes p -forms to depend on the differentials in ways appropriate (in three dimensions) for the description of elements of length, area, and volume, in part because the fact that $dx_i \wedge dx_i = 0$ prevents the appearance of duplicated differentials. In particular, 1-forms can be associated with elements of length, 2-forms with area, and 3-forms with volume. This feature carries forward to spaces of arbitrary dimensionality, thereby resolving potentially difficult questions that would otherwise have to be handled on a case-by-case basis. In fact, one of the virtues of the differential-forms approach is that there now exists a considerable body of general mathematical results that is pretty much completely absent from tensor analysis. For example, we will shortly find that the rules for differentiation in the exterior algebra cause the derivative of a p -form to be a $(p + 1)$ -form, thereby avoiding a pitfall that arises in tensor calculus: When the transformation coefficients are position-dependent, simply differentiating the coefficients representing a tensor of rank p does **not** yield another tensor. As we have seen, this dilemma is resolved in tensor analysis by introducing the notion of **covariant derivative**. Another consequence of the antisymmetry is that lengths, areas, volumes, and (at higher dimensionality) hypervolumes are **oriented** (meaning that they have signs that depend on the way the p -forms defining them are written), and the orientation must be taken into account when making computations based on differential forms.

Complementary Differential Forms

Associated with each differential form is a complementary (or **dual**) form that contains the differentials **not** included in the original form. Thus, if our underlying space has dimension d , the form dual to a p -form will be a $(d - p)$ -form. In three dimensions, the complement to a 1-form will be a 2-form (and vice versa), while the complement to a 3-form will be a 0-form (a scalar). It is useful to work with these complementary forms, and this is done by introducing an operator known as the **Hodge operator**; it is usually designated notationally as an asterisk (preceding the quantity to which it is applied, not as a superscript), and is therefore also referred to either as the **Hodge star operator** or simply as the **star operator**. Formally, its definition requires the introduction of a metric and the selection of an **orientation** (chosen by specifying the standard order of the differentials comprising the 1-form basis), and if the 1-form basis is not orthogonal there result complications we shall not discuss. For orthogonal bases, the dual forms depend on the index positions of the factors and on the metric tensor.⁶

⁶In the current discussion, restricted to Euclidean and Minkowski metrics, the metric tensor is diagonal, with diagonal elements ± 1 , and the relevant quantities are the signs of the diagonal elements.

To find $*\omega$, where ω is a p -form, we start by writing the wedge product ω' of all members of the 1-form basis not represented in ω , with the sign corresponding to the permutation that is needed to bring the index set

(indices of ω) followed by (indices of ω')

to standard order. Then $*\omega$ consists of ω' (with the sign we just found), but also multiplied by $(-1)^\mu$, where μ is the number of differentials in ω' whose metric-tensor diagonal element is -1 . For $\mathbb{R}^3$, ordinary 3-D space, the metric tensor is a unit matrix, so this final multiplication can be omitted, but it becomes relevant for our other case of current interest, the Minkowski metric.

For Euclidean 3-D space, we have

$$\begin{aligned} *1 &= dx_1 \wedge dx_2 \wedge dx_3, \\ *dx_1 &= dx_2 \wedge dx_3, \quad *dx_2 = dx_3 \wedge dx_1, \quad *dx_3 = dx_1 \wedge dx_2, \\ *(dx_1 \wedge dx_2) &= dx_3, \quad *(dx_3 \wedge dx_1) = dx_2, \quad *(dx_2 \wedge dx_3) = dx_1, \\ *(dx_1 \wedge dx_2 \wedge dx_3) &= 1. \end{aligned} \tag{4.81}$$

Cases not shown above are linearly dependent on those that were shown and can be obtained by permuting the differentials in the above formulas and taking the resulting sign changes into account.

At this point, two observations are in order. First, note that by writing the indices 1, 2, 3 in cyclic order, we have caused all the starred quantities to have positive signs. This choice makes the symmetry more evident. Second, it can be seen that all the formulas in Eq. (4.81) are consistent with $*(\omega) = \omega$. However, this is not universally true; compare with the formulas for Minkowski space, which are in the example we next consider. See also [Exercise 4.5.1](#).

Example 4.5.2 HODGE OPERATOR IN MINKOWSKI SPACE

Taking the oriented 1-form basis (dt, dx_1, dx_2, dx_3) , and the metric tensor

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix},$$

let's determine the effect of the Hodge operator on the various possible differential forms. Consider initially $*1$, for which the complementary form contains $dt \wedge dx_1 \wedge dx_2 \wedge dx_3$. Since we took these differentials in the basis order, they are assigned a plus sign. Since $\omega = 1$ contains no differentials, its number μ of negative metric-tensor diagonal elements is zero, so $(-1)^\mu = (-1)^0 = 1$ and there is no sign change arising from the metric. Therefore,

$$*1 = dt \wedge dx_1 \wedge dx_2 \wedge dx_3.$$

Next, take $*(dt \wedge dx_1 \wedge dx_2 \wedge dx_3)$. The complementary form is just unity, with no sign change due to the index ordering, as the differentials are already in standard order.

However, this time we have three entries in the quantity being starred with negative metric-tensor diagonal elements; this generates $(-1)^3 = -1$, so

$$*(dt \wedge dx_1 \wedge dx_2 \wedge dx_3) = -1.$$

Moving next to $*dx_1$, the complementary form is $dt \wedge dx_2 \wedge dx_3$, and the index ordering (based on dx_1, dt, dx_2, dx_3) requires one pair interchange to reach the standard order (thereby yielding a minus sign). But the quantity being starred contains one differential that generates a minus sign, namely dx_1 , so

$$*dx_1 = dt \wedge dx_2 \wedge dx_3.$$

Looking explicitly at one more case, consider $*(dt \wedge dx_1)$, for which the complementary form is $dx_2 \wedge dx_3$. This time the indices are in standard order, but the dx_1 being starred generates a minus sign, so

$$*(dt \wedge dx_1) = -dx_2 \wedge dx_3.$$

Development of the remaining possibilities is left to [Exercise 4.5.1](#); the results are summarized below, where i, j, k denotes any cyclic permutation of 1, 2, 3.

$$\begin{aligned} *1 &= dt \wedge dx_1 \wedge dx_2 \wedge dx_3, \\ *dx_i &= dt \wedge dx_j \wedge dx_k, \quad *dt = dx_1 \wedge dx_2 \wedge dx_3, \\ *(dx_j \wedge dx_k) &= dt \wedge dx_i, \quad *(dt \wedge dx_i) = -dx_j \wedge dx_k, \\ *(dx_1 \wedge dx_2 \wedge dx_3) &= dt, \quad *(dt \wedge dx_i \wedge dx_j) = dx_k, \\ *(dt \wedge dx_1 \wedge dx_2 \wedge dx_3) &= -1. \end{aligned} \tag{4.82}$$

Note that all the starred forms in [Eq. \(4.82\)](#) with an even number of differentials have the property that $*(\omega) = -\omega$, confirming our earlier statement that complementing twice does not always restore the original form with its original sign. ■

We now consider some examples illustrating the utility of the star operator.

Example 4.5.3 MISCELLANEOUS DIFFERENTIAL FORMS

In the Euclidean space $\mathbb{R}^3$, consider the wedge product $A \wedge B$ of the two 1-forms $A = A_x dx + A_y dy + A_z dz$ and $B = B_x dx + B_y dy + B_z dz$. Simplifying using the rules for exterior products,

$$A \wedge B = (A_y B_z - A_z B_y) dy \wedge dz + (A_z B_x - A_x B_z) dz \wedge dx + (A_x B_y - A_y B_x) dx \wedge dy.$$

If we now apply the star operator and use the formulas in [Eq. \(4.81\)](#) we get

$$*(A \wedge B) = (A_y B_z - A_z B_y) dx + (A_z B_x - A_x B_z) dy + (A_x B_y - A_y B_x) dz,$$

showing that in $\mathbb{R}^3$, $*(A \wedge B)$ forms an expression that is analogous to the cross product $\mathbf{A} \times \mathbf{B}$ of vectors $A_x \hat{\mathbf{e}}_x + A_y \hat{\mathbf{e}}_y + A_z \hat{\mathbf{e}}_z$ and $B_x \hat{\mathbf{e}}_x + B_y \hat{\mathbf{e}}_y + B_z \hat{\mathbf{e}}_z$. In fact, we can write

$$*(A \wedge B) = (\mathbf{A} \times \mathbf{B})_x dx + (\mathbf{A} \times \mathbf{B})_y dy + (\mathbf{A} \times \mathbf{B})_z dz. \tag{4.83}$$

Note that the sign of $*(A \wedge B)$ is determined by our implicit choice that the standard ordering of the basis differentials is (dx, dy, dz) .

Next, consider the exterior product $A \wedge B \wedge C$, where C is a 1-form with coefficients C_x, C_y, C_z . Applying the evaluation rules, we find that every surviving term in the product is proportional to $dx \wedge dy \wedge dz$, and we obtain

$$\begin{aligned} A \wedge B \wedge C = & (A_x B_y C_z - A_x B_z C_y - A_y B_x C_z \\ & + A_y B_z C_x + A_z B_x C_y - A_z B_y C_x) dx \wedge dy \wedge dz, \end{aligned}$$

which we recognize can be written in the form

$$A \wedge B \wedge C = \begin{vmatrix} A_x & A_y & A_z \\ B_x & B_y & B_z \\ C_x & C_y & C_z \end{vmatrix} dx \wedge dy \wedge dz.$$

Applying now the star operator, we reach

$$*(A \wedge B \wedge C) = \begin{vmatrix} A_x & A_y & A_z \\ B_x & B_y & B_z \\ C_x & C_y & C_z \end{vmatrix} = \mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}). \quad (4.84)$$

Not only were the results in Eqs. (4.83) and (4.84) easily obtained, they also generalize nicely to spaces of arbitrary dimension and metric, while the traditional vector notation, which uses the cross product, is applicable only to $\mathbb{R}^3$. ■

Exercises

- 4.5.1** Using the rules for the application of the Hodge star operator, verify the results given in Eq. (4.82) for its application to all linearly independent differential forms in Minkowski space.
- 4.5.2** If the force field is constant and moving a particle from the origin to $(3, 0, 0)$ requires a units of work, from $(-1, -1, 0)$ to $(-1, 1, 0)$ takes b units of work, and from $(0, 0, 4)$ to $(0, 0, 5)$ c units of work, find the 1-form of the work.

4.6 DIFFERENTIATING FORMS

Exterior Derivatives

Having introduced differential forms and their exterior algebra, we next develop their properties under differentiation. To accomplish this, we define the **exterior derivative**, which we consider to be an **operator** identified by the traditional symbol d . We have, in fact, already introduced that operator when we wrote dx_i , stating at the time that we intended to interpret dx_i as a mathematical object with specified properties and not just as a small

change in x_i . We are now refining that statement to interpret dx_i as the result of applying the operator d to the quantity x_i . We complete our definition of the operator d by requiring it to have the following properties, where ω is a p -form, ω' is a p' -form, and f is an ordinary function (a 0-form):

$$\begin{aligned}
 d(\omega + \omega') &= d\omega + d\omega' \quad (p = p'), \\
 d(f\omega) &= (df) \wedge \omega + f d\omega, \\
 d(\omega \wedge \omega') &= d\omega \wedge \omega' + (-1)^p \omega \wedge d\omega', \\
 d(d\omega) &= 0, \\
 df &= \sum_j \frac{\partial f}{\partial x_j} dx_j,
 \end{aligned} \tag{4.85}$$

where the sum over j spans the underlying space. The formula for the derivative of the wedge product is sometime called by mathematicians an **antiderivation**, referring to the fact that when applied to the right-hand factor an antisymmetry-motivated minus sign appears.

Example 4.6.1 EXTERIOR DERIVATIVE

Equations (4.85) are *axioms*, so they are not subject to proof, though they **are** required to be consistent. It is of interest to verify that the sign for the derivative of the second term in a wedge product is needed. Taking ω and ω' to be monomials, we first bring their coefficients to the left and then apply the differentiation operator (which, irrespective of the choice of sign, gives zero when applied to any of the differentials). Thus,

$$\begin{aligned}
 d(\omega \wedge \omega') &= d(AB) \left[dx_1 \wedge \cdots \wedge dx_p \right] \wedge \left[dx_1 \wedge \cdots \wedge dx_{p'} \right] \\
 &= \sum_{\mu} \left[\frac{\partial A}{\partial x_{\mu}} B + A \frac{\partial B}{\partial x_{\mu}} \right] dx_{\mu} \wedge \left[dx_1 \wedge \cdots \wedge dx_p \right] \wedge \left[dx_1 \wedge \cdots \wedge dx_{p'} \right].
 \end{aligned}$$

On expanding the sum, the first term is clearly $d\omega \wedge \omega'$; to make the second term look like $\omega \wedge d\omega'$, it is necessary to permute dx_{μ} through the p differentials in ω , yielding the sign factor $(-1)^p$. Extension to general polynomial forms is trivial.

One might also ask whether the fourth of the above axioms, $d(d\omega) = 0$, sometimes referred to as **Poincaré's lemma**, is necessary or consistent with the others. First, it provides new information, as otherwise we have no way of reducing $d(dx_i)$. Next, to see why the axiom set is consistent, we illustrate by examining (in $\mathbb{R}^2$)

$$df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy,$$

from which we form

$$\begin{aligned} d(df) &= \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} \right) dx \wedge dx + \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right) dy \wedge dx \\ &\quad + \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right) dx \wedge dy + \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial y} \right) dy \wedge dy = 0. \end{aligned}$$

We obtain the zero result because of the antisymmetry of the wedge product and because the mixed second derivatives are equal. We see that the central reason for the validity of Poincaré's lemma is that the mixed derivatives of a sufficiently differentiable function are invariant with respect to the order in which the differentiations are carried out. ■

To catalog the possibilities for the action of the d operator in ordinary 3-D space, we first note that the derivative of an ordinary function (a 0-form) is

$$df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy + \frac{\partial f}{\partial z} dz = (\nabla f)_x dx + (\nabla f)_y dy + (\nabla f)_z dz. \quad (4.86)$$

We next differentiate the 1-form $\omega = A_x dx + A_y dy + A_z dz$. After simplification,

$$d\omega = \left[\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right] dy \wedge dz + \left[\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right] dz \wedge dx + \left[\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right] dx \wedge dy.$$

We recognize this as

$$\begin{aligned} d(A_x dx + A_y dy + A_z dz) &= \\ &= (\nabla \times \mathbf{A})_x dy \wedge dz + (\nabla \times \mathbf{A})_y dz \wedge dx + (\nabla \times \mathbf{A})_z dx \wedge dy, \end{aligned} \quad (4.87)$$

which is equivalent to

$$*d(A_x dx + A_y dy + A_z dz) = (\nabla \times \mathbf{A})_x dx + (\nabla \times \mathbf{A})_y dy + (\nabla \times \mathbf{A})_z dz. \quad (4.88)$$

Finally we differentiate the 2-form $B_x dy \wedge dz + B_y dz \wedge dx + B_z dx \wedge dy$, obtaining the three-form

$$d(B_x dy \wedge dz + B_y dz \wedge dx + B_z dx \wedge dy) = \left[\frac{\partial B_x}{\partial x} + \frac{\partial B_y}{\partial y} + \frac{\partial B_z}{\partial z} \right] dx \wedge dy \wedge dz,$$

equivalent to

$$d(B_x dy \wedge dz + B_y dz \wedge dx + B_z dx \wedge dy) = (\nabla \cdot \mathbf{B}) dx \wedge dy \wedge dz \quad (4.89)$$

and

$$*d(B_x dy \wedge dz + B_y dz \wedge dx + B_z dx \wedge dy) = \nabla \cdot \mathbf{B}. \quad (4.90)$$

We see that application of the d operator directly generates all the differential operators of traditional vector analysis.

If now we return to Eq. (4.87) and take the 1-form on its left-hand side to be df , so that $\mathbf{A} = \nabla f$, we have, inserting Eq. (4.86),

$$d(df) = \left(\nabla \times (\nabla f) \right)_x dy \wedge dz + \left(\nabla \times (\nabla f) \right)_y dz \wedge dx + \left(\nabla \times (\nabla f) \right)_z dx \wedge dy = 0. \quad (4.91)$$

We have invoked Poincaré's lemma to set this expression to zero. The result is equivalent to the well-known identity $\nabla \times (\nabla f) = 0$.

Another identity is obtained if we start from Eq. (4.89) and take the 2-form on its left-hand side to be $d(A_x dx + A_y dy + A_z dz)$. Then, with the aid of Eq. (4.88), we have

$$d\left(d(A_x dx + A_y dy + A_z dz)\right) = \nabla \cdot (\nabla \times \mathbf{A}) dx \wedge dy \wedge dz = 0, \quad (4.92)$$

where once again the zero result follows from Poincaré's lemma and we have established the well-known formula $\nabla \cdot (\nabla \times \mathbf{A}) = 0$. Part of the importance of the derivation of these formulas using differential-forms methods is that these are merely the first members of hierarchies of identities that can be derived for spaces with higher numbers of dimensions and with different metric properties.

Example 4.6.2 MAXWELL'S EQUATIONS

Maxwell's equations of electromagnetic theory can be written in an extremely compact and elegant way using differential forms notation. In that notation, the independent elements of the electromagnetic field tensor can be written as the coefficients of a 2-form in Minkowski space with oriented basis (dt, dx, dy, dz) :

$$\begin{aligned} F = & -E_x dt \wedge dx - E_y dt \wedge dy - E_z dt \wedge dz \\ & + B_x dy \wedge dz + B_y dz \wedge dx + B_z dx \wedge dy. \end{aligned} \quad (4.93)$$

Here $\mathbf{E}$ and $\mathbf{B}$ are respectively the electric field and the magnetic induction. The sources of the field, namely the charge density ρ and the components of the current density $\mathbf{J}$, become the coefficients of the 3-form

$$J = \rho dx \wedge dy \wedge dz - J_x dt \wedge dy \wedge dz - J_y dt \wedge dz \wedge dx - J_z dt \wedge dx \wedge dy. \quad (4.94)$$

For simplicity we work in units with the permittivity, magnetic permeability, and velocity of light all set to unity ($\varepsilon = \mu = c = 1$). Note that it is natural that the charge and current densities occur in a 3-form; although they have together the number of components needed to constitute a four-vector, they are of dimension inverse volume. Note also that some of the signs in the formulas of this example depend on the details of the metric, and are chosen to be correct for the Minkowski metric as given in Example 4.5.2. This Minkowski metric has **signature** (1,3), meaning that it has one positive and three negative diagonal elements. Some workers define the Minkowski metric to have signature (3,1), reversing all its signs. Either choice will give correct results to problems of physics if used consistently; trouble only arises if material from inconsistent sources is combined.

The two homogeneous Maxwell equations are obtained from the simple formula $dF = 0$. This equation is not a mathematical requirement on F ; it is a statement of the

physical properties of electric and magnetic fields. To relate our new formula to the more usual vector equations, we simply apply the d operator to F :

$$\begin{aligned} dF = & - \left[\frac{\partial E_x}{\partial y} dy + \frac{\partial E_x}{\partial z} dz \right] \wedge dt \wedge dx - \left[\frac{\partial E_y}{\partial x} dx + \frac{\partial E_y}{\partial z} dz \right] \wedge dt \wedge dy \\ & - \left[\frac{\partial E_z}{\partial x} dx + \frac{\partial E_z}{\partial y} dy \right] \wedge dt \wedge dz + \left[\frac{\partial B_x}{\partial t} dt + \frac{\partial B_x}{\partial x} dx \right] \wedge dy \wedge dz \\ & + \left[\frac{\partial B_y}{\partial t} dt + \frac{\partial B_y}{\partial y} dy \right] \wedge dz \wedge dx + \left[\frac{\partial B_z}{\partial t} dt + \frac{\partial B_z}{\partial z} dz \right] \wedge dx \wedge dy = 0. \end{aligned} \quad (4.95)$$

Equation (4.95) is easily simplified to

$$\begin{aligned} dF = & \left[\frac{\partial E_z}{\partial y} - \frac{\partial E_y}{\partial z} + \frac{\partial B_x}{\partial t} \right] dt \wedge dy \wedge dz + \left[\frac{\partial E_x}{\partial z} - \frac{\partial E_z}{\partial x} + \frac{\partial B_y}{\partial t} \right] dt \wedge dz \wedge dx \\ & + \left[\frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y} + \frac{\partial B_z}{\partial t} \right] dt \wedge dx \wedge dy + \left[\frac{\partial B_x}{\partial x} + \frac{\partial B_y}{\partial y} + \frac{\partial B_z}{\partial z} \right] dx \wedge dy \wedge dz = 0. \end{aligned} \quad (4.96)$$

Since the coefficient of each 3-form monomial must individually vanish, we obtain from Eq. (4.96) the vector equations

$$\nabla \times \mathbf{E} + \frac{\partial \mathbf{B}}{\partial t} = 0 \quad \text{and} \quad \nabla \cdot \mathbf{B} = 0.$$

We now go on to obtain the two inhomogeneous Maxwell equations from the almost equally simple formula $d(*F) = J$. To verify this, we first form $*F$, evaluating the starred quantities using the formulas in Eqs. (4.82):

$$*F = E_x dy \wedge dz + E_y dz \wedge dx + E_z dx \wedge dy + B_x dt \wedge dx + B_y dt \wedge dy + B_z dt \wedge dz.$$

We now apply the d operator, reaching after steps similar to those taken while obtaining Eq. (4.96):

$$\begin{aligned} d(*F) = & \nabla \cdot \mathbf{E} dx \wedge dy \wedge dz + \left[\frac{\partial E_x}{\partial t} - (\nabla \times \mathbf{B})_x \right] dt \wedge dy \wedge dz \\ & + \left[\frac{\partial E_y}{\partial t} - (\nabla \times \mathbf{B})_y \right] dt \wedge dz \wedge dx + \left[\frac{\partial E_z}{\partial t} - (\nabla \times \mathbf{B})_z \right] dt \wedge dx \wedge dy. \end{aligned} \quad (4.97)$$

Setting $d(*F)$ from Eq. (4.97) equal to J as given in Eq. (4.94), we obtain the remaining Maxwell equations

$$\nabla \cdot \mathbf{E} = \rho \quad \text{and} \quad \nabla \times \mathbf{B} - \frac{\partial \mathbf{E}}{\partial t} = \mathbf{J}.$$

We close this example by applying the d operator to J . The result must vanish because $dJ = d(d(*F))$. We get, starting from Eq. (4.94),

$$dJ = \left[\frac{\partial \rho}{\partial t} + \frac{\partial J_x}{\partial x} + \frac{\partial J_y}{\partial y} + \frac{\partial J_z}{\partial z} \right] dt \wedge dx \wedge dy \wedge dz = 0,$$

showing that

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{J} = 0. \quad (4.98)$$

Summarizing, the differential-forms approach has reduced Maxwell's equations to the two simple formulas

$$dF = 0 \quad \text{and} \quad d(*F) = J, \quad (4.99)$$

and we have also shown that $\mathbf{J}$ must satisfy an equation of continuity. ■

Exercises

- 4.6.1** Given the two 1-forms $\omega_1 = x dy + y dx$ and $\omega_2 = x dy - y dx$, calculate
- (a) $d\omega_1$,
 - (b) $d\omega_2$.
 - (c) For each of your answers to (a) or (b) that is nonzero, apply the operator d a second time and verify that $d(d\omega_i) = 0$.
- 4.6.2** Apply the operator d twice to $\omega_3 = xy dz + xz dy - yz dx$. Verify that the second application of d yields a zero result.
- 4.6.3** For ω_2 and ω_3 the 1-forms with these names in [Exercises 4.6.1](#) and [4.6.2](#), evaluate $d(\omega_2 \wedge \omega_3)$:
- (a) By forming the exterior product and then differentiating, and
 - (b) Using the formula for differentiating a product of two forms.

Verify that both approaches give the same result.

4.7 INTEGRATING FORMS

It is natural to define the integrals of differential forms in a way that preserves our usual notions of integration. The integrals with which we are concerned are over regions of the manifolds on which our differential forms are defined; this fact and the antisymmetry of the wedge product need to be taken into account in developing definitions and properties of integrals. For convenience, we illustrate in two or three dimensions; the notions extend to spaces of arbitrary dimensionality.

Consider first the integral of a 1-form ω in 2-D space, integrated over a curve C from a start-point P to an endpoint Q :

$$\int_C \omega = \int_C [A_x dx + A_y dy].$$

We interpret the integration as a conventional line integral. If the curve is described parametrically by $x(t)$, $y(t)$ as t increases monotonically from t_P to t_Q , our integral takes the

elementary form

$$\int_C \omega = \int_{t_P}^{t_Q} \left[A_x(t) \frac{dx}{dt} + A_y(t) \frac{dy}{dt} \right] dt,$$

and (at least in principle) the integral can be evaluated by the usual methods.

Sometimes the integral will have a value that will be independent of the path from P to Q ; in physics this situation arises when a 1-form with coefficients $\mathbf{A} = (A_x, A_y)$ describes what is known as a **conservative force** (i.e., one that can be written as the gradient of a potential). In our present language, we then call ω **exact**, meaning that there exists some function f such that

$$\omega = df(x, y) \quad (4.100)$$

for a region that includes the points P , Q , and all other points through which the path may pass.

To check the significance of Eq. (4.100), note that it implies

$$\omega = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy,$$

showing that ω has as coefficients the components of the gradient of f . Given Eq. (4.100), we also see that

$$\text{if } \omega = df, \quad \int_P^Q \omega = f(Q) - f(P). \quad (4.101)$$

This admittedly obvious result is independent of the dimension of the space, and is of importance to the remainder of this section.

Looking next at 2-forms, we have (in 2-D space) integrals such as

$$\int_S \omega = \int_S B(x, y) dx \wedge dy. \quad (4.102)$$

We interpret $dx \wedge dy$ as the element of area corresponding to displacements dx and dy in mutually orthogonal directions, so in the usual notation of integral calculus we would write $dx dy$.

Let's now return to the wedge product notation and consider what happens if we make a change of variables from x, y to u, v , with $x = au + bv$, $y = eu + fv$. Then $dx = a du + b dv$, $dy = e du + f dv$, and

$$dx \wedge dy = (a du + b dv) \wedge (e du + f dv) = (af - be) du \wedge dv. \quad (4.103)$$

We note that the coefficient of $du \wedge dv$ is just the Jacobian of the transformation from x, y to u, v , which becomes clear if we write $a = \partial x / \partial u$, etc., after which we have

$$af - be = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} a & b \\ e & f \end{vmatrix}. \quad (4.104)$$

We now see a fundamental reason why the wedge product has been introduced; it has the algebraic properties needed to generate in a natural fashion the relations between elements of area (or its higher-dimension analogs) in different coordinate systems. To emphasize that observation, note that the Jacobian occurred as a natural consequence of the transformation; we did not have to take additional steps to insert it, and it was generated simply by evaluating the relevant differential forms. In addition, the present formulation has one new feature: because $dx \wedge dy$ and $dy \wedge dx$ are opposite in sign, areas must be assigned algebraic signs, and it is necessary to retain the sign of the Jacobian if we make a change of variables. We therefore take as the element of area corresponding to $dx \wedge dy$ the ordinary product $\pm dx dy$, with a choice of sign known as the **orientation** of the area. Then, Eq. (4.102) becomes

$$\int_S \omega = \int_S B(x, y)(\pm dx dy), \quad (4.105)$$

and if elsewhere in the same computation we had $dy \wedge dx$, we must convert it to $dx dy$ using the sign opposite to that used for $dx \wedge dy$.

For p -forms with $p > 2$, a corresponding analysis applies: If we transform from $(x, y, \dots)$ to $(u, v, \dots)$, the wedge product $dx \wedge dy \wedge \dots$ becomes $J du \wedge dv \wedge \dots$, where J is the (signed) Jacobian of the transformation. Since the p -space volumes are *oriented*, the sign of the Jacobian is relevant and must be retained. Exercise 4.7.1 shows that the change of variables from the 3-form $dx \wedge dy \wedge dz$ to $du \wedge dv \wedge dw$ yields the determinant which is the (signed) Jacobian of the transformation.

Stokes' Theorem

A key result regarding the integration of differential forms is a formula known as **Stokes' theorem**, a restricted form of which we encountered in our study of vector analysis in Chapter 3. Stokes' theorem, in its simplest form, states that if

- R is a simply-connected region (i.e., one with no holes) of a p -dimensional differentiable manifold in a n -dimensional space ($n \geq p$);
- R has a boundary denoted ∂R , of dimension $p - 1$;
- ω is a $(p - 1)$ -form defined on R and its boundary, with derivative $d\omega$;

then

$$\int_R d\omega = \int_{\partial R} \omega. \quad (4.106)$$

This is the generalization, to p dimensions, of Eq. (4.101). Note that because $d\omega$ results from applying the d operator to ω , the differentials in $d\omega$ consist of all those in ω , in the same order, but preceded by that produced by the differentiation. This observation is relevant for identifying the signs to be associated with the integrations.

A rigorous proof of Stokes' theorem is somewhat complicated, but an indication of its validity is not too involved. It is sufficient to consider the case that ω is a monomial:

$$\omega = A(x_1, \dots, x_p) dx_2 \wedge \dots \wedge dx_p, \quad d\omega = \frac{\partial A}{\partial x_1} dx_1 \wedge dx_2 \wedge \dots \wedge dx_p. \quad (4.107)$$

We start by approximating the portion of R adjacent to the boundary by a set of small p -dimensional parallelepipeds whose thickness in the x_1 direction is δ , with δ having for each parallelepiped the sign that makes $x_1 \rightarrow x_1 - \delta$ in the interior of R . For each such parallelepiped (symbolically denoted Δ , with faces of constant x_1 denoted $\partial\Delta$), we integrate $d\omega$ in x_1 from $x_1 - \delta$ to x_1 and over the full range of the other x_i , obtaining

$$\begin{aligned} \int_{\Delta} d\omega &= \int_{\partial\Delta} \int_{x_1-\delta}^{x_1} \left(\frac{\partial A}{\partial x_1} \right) dx_1 \wedge dx_2 \wedge \dots \wedge dx_p \\ &= \int_{\partial\Delta} A(x_1, x_2, \dots) dx_2 \wedge \dots \wedge dx_p - \int_{\partial\Delta} A(x_1 - \delta, x_2, \dots) dx_2 \wedge \dots \wedge dx_p. \end{aligned} \quad (4.108)$$

Equation (4.108) indicates the validity of Stokes' theorem for a laminar region whose exterior boundary is ∂R ; if we perform the same process repeatedly, we can collapse the inner boundary to a region of zero volume, thereby reaching Eq. (4.106).

Stokes' theorem applies for manifolds of any dimension; different cases of this single theorem in two and three dimensions correspond to results originally identified as distinct theorems. Some examples follow.

Example 4.7.1 GREEN'S THEOREM IN THE PLANE

Consider in a 2-D space the 1-form ω and its derivative:

$$\omega = P(x, y) dx + Q(x, y) dy, \quad (4.109)$$

$$d\omega = \frac{\partial P}{\partial y} dy \wedge dx + \frac{\partial Q}{\partial x} dx \wedge dy = \left[\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right] dx \wedge dy, \quad (4.110)$$

where we have without comment discarded terms containing $dx \wedge dx$ or $dy \wedge dy$.

We apply Stokes' theorem for this ω to a region S with boundary C , obtaining

$$\int_S \left[\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right] dx \wedge dy = \int_C (P dx + Q dy).$$

With orientation such that $dx \wedge dy = dS$ (ordinary element of area), we have the formula usually identified as *Green's theorem in the plane*:

$$\int_C (P dx + Q dy) = \int_S \left[\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right] dS. \quad (4.111)$$

Some cases of this theorem: taking $P = 0$, $Q = x$, we have the well-known formula

$$\int_C x dy = \int_S dS = A,$$

where A is the area enclosed by C with the line integral evaluated in the mathematically positive (counterclockwise) direction.

If we take $P = y$, $Q = 0$, we get instead another familiar formula:

$$\int_C y dx = \int_S (-1) dS = -A.$$

■

When working [Example 4.7.1](#), we assumed (without comment) that the line integral on the closed curve C was to be evaluated for travel in the counterclockwise direction, and we also related area to the conversion from $dx \wedge dy$ to $+dx dy$. These are choices that were not dictated by the theory of differential forms but by our intention to make its results correspond to computation in the usual system of planar Cartesian coordinates. What is certainly true is that the differential forms calculus gives a different sign for the integral of $y dx$ than it gave for the integral of $x dy$; the user of the calculus has the responsibility to make definitions corresponding to the situation for which the results are claimed to be relevant.

Example 4.7.2 STOKES' THEOREM (USUAL 3-D CASE)

Let the vector potential $\mathbf{A}$ be represented by the differential form ω , with it and its derivative of the forms

$$\omega = A_x dx + A_y dy + A_z dz, \quad (4.112)$$

$$\begin{aligned} d\omega &= \left[\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right] dy \wedge dz + \left[\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right] dz \wedge dx + \left[\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right] dx \wedge dy \\ &= (\nabla \times \mathbf{A})_x dy \wedge dz + (\nabla \times \mathbf{A})_y dz \wedge dx + (\nabla \times \mathbf{A})_z dx \wedge dy. \end{aligned} \quad (4.113)$$

Applying Stokes' theorem to a region S with boundary C and noting that if the standard order for orienting the differentials is dx, dy, dz , then $dy \wedge dz \rightarrow d\sigma_x$, $dz \wedge dx \rightarrow d\sigma_y$, $dx \wedge dy \rightarrow d\sigma_z$, and Stokes' theorem takes the familiar form

$$\int_C (A_x dx + A_y dy + A_z dz) = \int_C \mathbf{A} \cdot d\mathbf{r} = \int_S (\nabla \times \mathbf{A}) \cdot d\boldsymbol{\sigma}. \quad (4.114)$$

■

Once again we have results whose interpretation depends on how we have chosen to define the quantities involved. The differential forms calculus does not know whether we intend to use a right-handed coordinate system, and that choice is implicit in our identification of the elements of area $d\sigma_j$. In fact, the mathematics does not even tell us that the quantities we identified as components of $\nabla \times \mathbf{A}$ actually correspond to anything physical

in their indicated directions. So, once again, we emphasize that the mathematics of differential forms provides a structure appropriate to the physics to which we apply it, but part of what the physicist brings to the table is the correlation between mathematical objects and the physical quantities they represent.

Example 4.7.3 GAUSS' THEOREM

As a final example, consider a 3-D region V with boundary ∂V , containing an electric field given on ∂V as the 2-form ω , with

$$\omega = E_x dy \wedge dz + E_y dz \wedge dx + E_z dx \wedge dy, \quad (4.115)$$

$$d\omega = \left[\frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} + \frac{\partial E_z}{\partial z} \right] dx \wedge dy \wedge dz = (\nabla \cdot \mathbf{E}) dx \wedge dy \wedge dz. \quad (4.116)$$

For this case, Stokes' theorem is

$$\int_V d\omega = \int_V (\nabla \cdot \mathbf{E}) dx \wedge dy \wedge dz = \int_V (\nabla \cdot \mathbf{E}) d\tau = \int_{\partial V} \mathbf{E} \cdot d\boldsymbol{\sigma}, \quad (4.117)$$

where $dx \wedge dy \wedge dz \rightarrow d\tau$ and, just as in Example 4.7.2, $dy \wedge dz \rightarrow d\sigma_x$, etc. We have recovered Gauss' theorem. ■

Exercises

4.7.1 Use differential-forms relations to transform the integral $A(x, y, z) dx \wedge dy \wedge dz$ to the equivalent expression in $du \wedge dv \wedge dw$, where u, v, w is a linear transformation of x, y, z , and thereby find the determinant that can be identified as the Jacobian of the transformation.

4.7.2 Write Oersted's law,

$$\int_{\partial S} \mathbf{H} \cdot d\mathbf{r} = \int_S \nabla \times \mathbf{H} \cdot d\mathbf{a} \sim I,$$

in differential form notation.

4.7.3 A 1-form $A dx + B dy$ is defined as **closed** if $\frac{\partial A}{\partial y} = \frac{\partial B}{\partial x}$. It is called **exact** if there is a

function f such that $\frac{\partial f}{\partial x} = A$ and $\frac{\partial f}{\partial y} = B$. Determine which of the following 1-forms are closed, or exact, and find the corresponding functions f for those that are exact:

$$y dx + x dy, \quad \frac{y dx + x dy}{x^2 + y^2}, \quad [\ln(xy) + 1] dx + \frac{x}{y} dy, \\ -\frac{y dx}{x^2 + y^2} + \frac{x dy}{x^2 + y^2}, \quad f(z) dz \text{ with } z = x + iy.$$

Additional Readings

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- Young, E. C., *Vector and Tensor Analysis*, 2nd ed. New York: Dekker (1993).